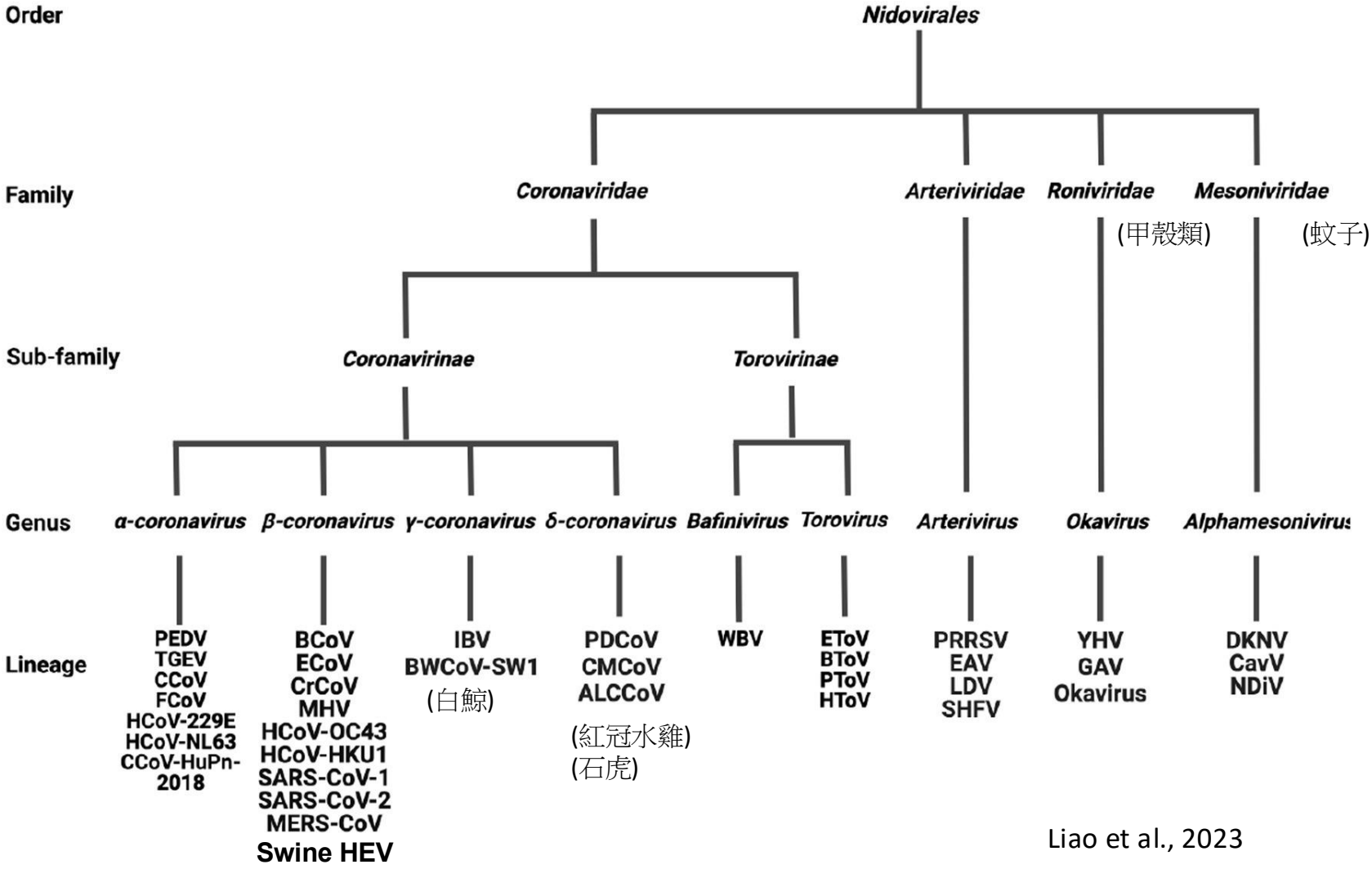
Swine Coronaviral Diseases

Chi-Fei Kao (高啓霏)

2026.10.08

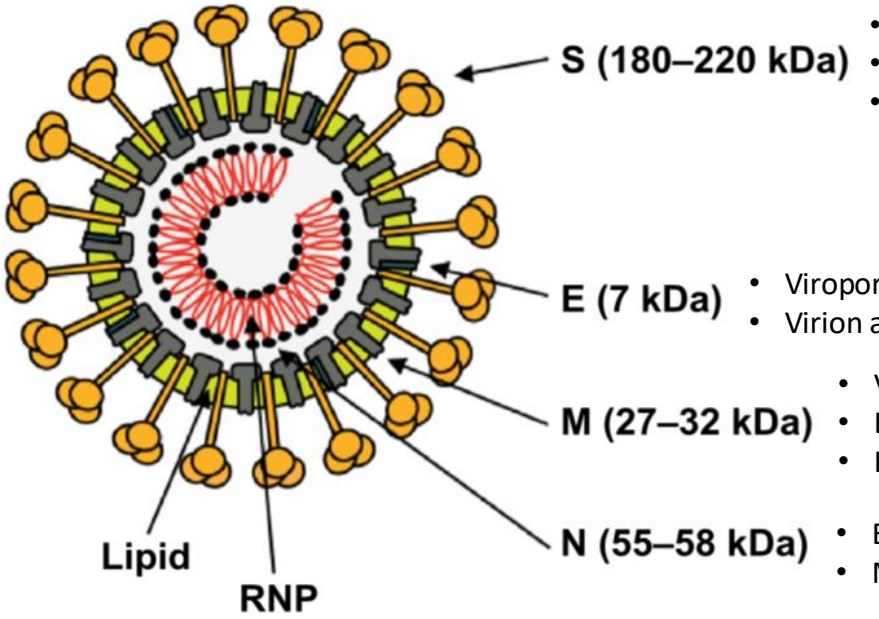
Coronaviruses



Liao et al., 2023

Features of Coronaviridae

- Characteristic virion morphology
  - Spherical to pleomorphic, enveloped viral particles (~100 nm in diameter) with S proteins on the surface.

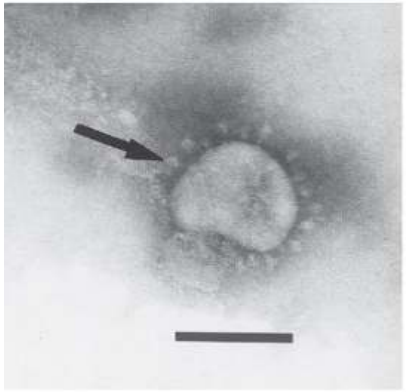


- Receptor binding
- Membrane fusion
- Induction of neutralizing antibodies

- Viroporin
- Virion assembly

- Viral morphogenesis
- Elicit protective immunity
- Participate in host-cell interaction

- Encapsulate viral RNA
- Mediate polymerase template switch



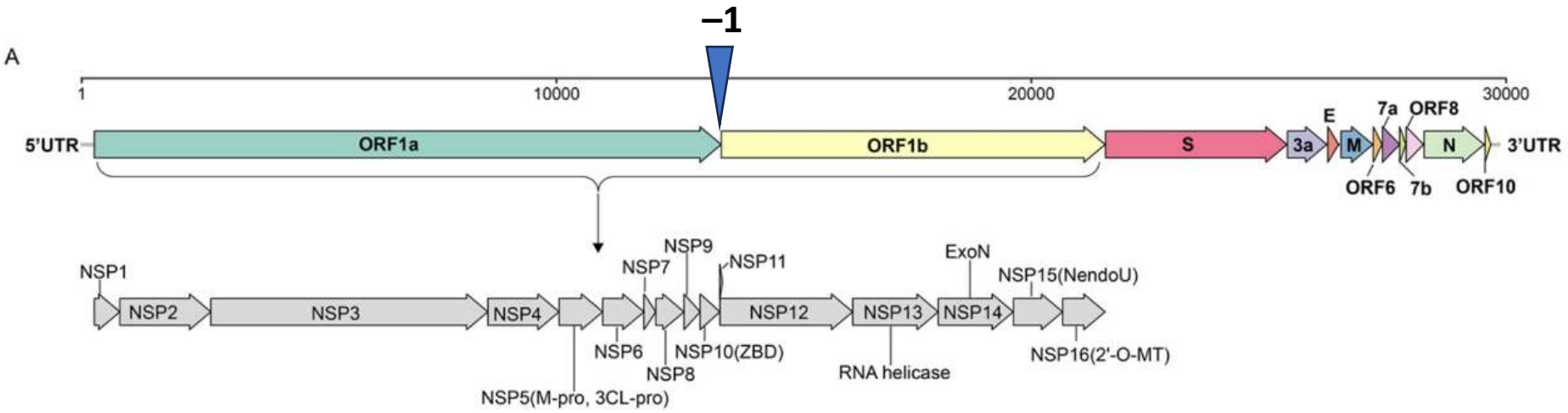
Lee et al., 2015

Diseases of Swine, 11th Edition

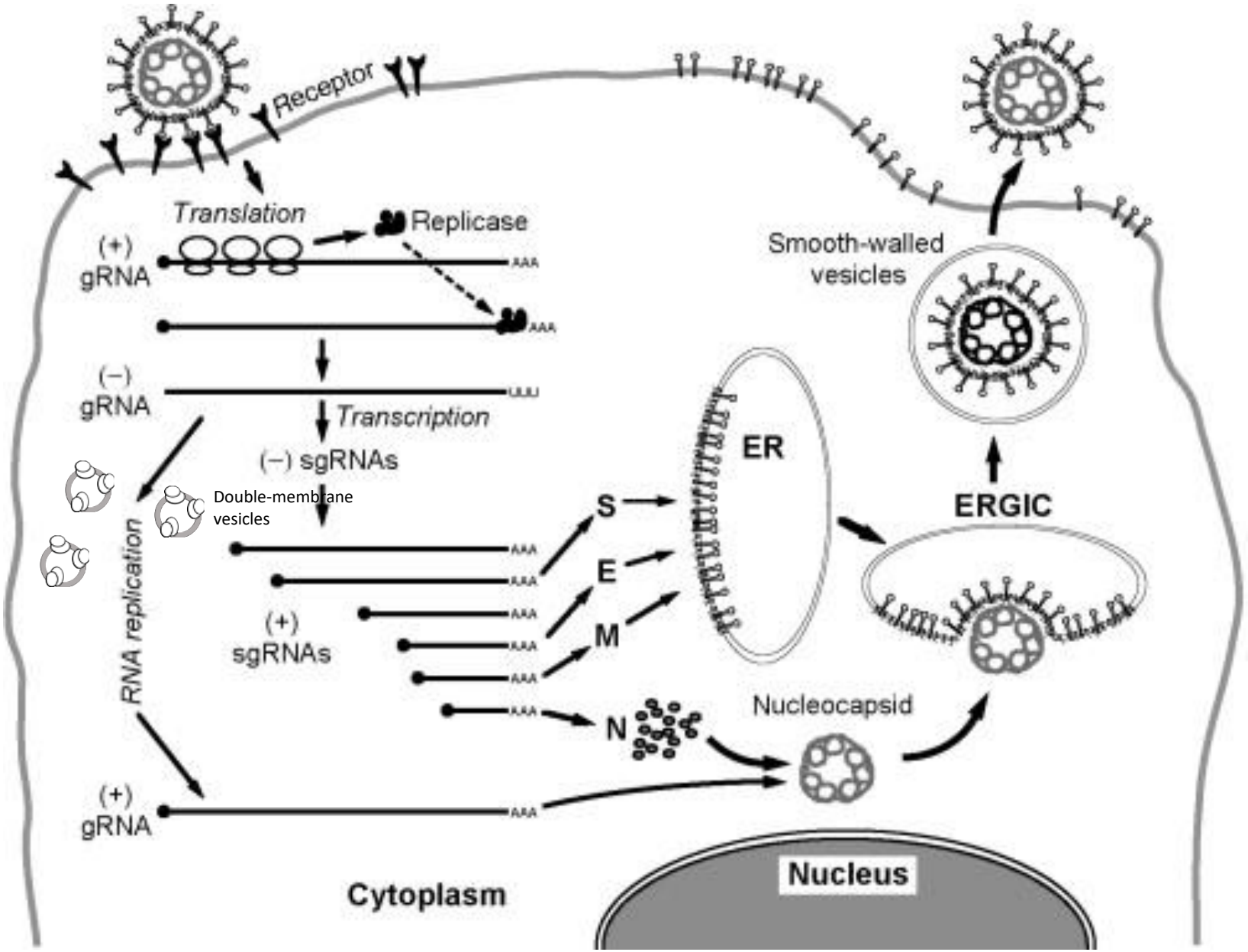
Features of Coronaviridae

- During viral replication:
  - ✓Similar genome organization
  - ✓Transcription of multiple **3'-nested subgenomic RNAs**
  - ✓Expression of the polyprotein by **–1 ribosomal frameshift**

Cross-reactivity with the N protein has also been reported for PEDV, FIPV, CCoV, TGEV, and a putative mink CoV.



Modified from Zhou et al,2021



Modified from Masters, 2006

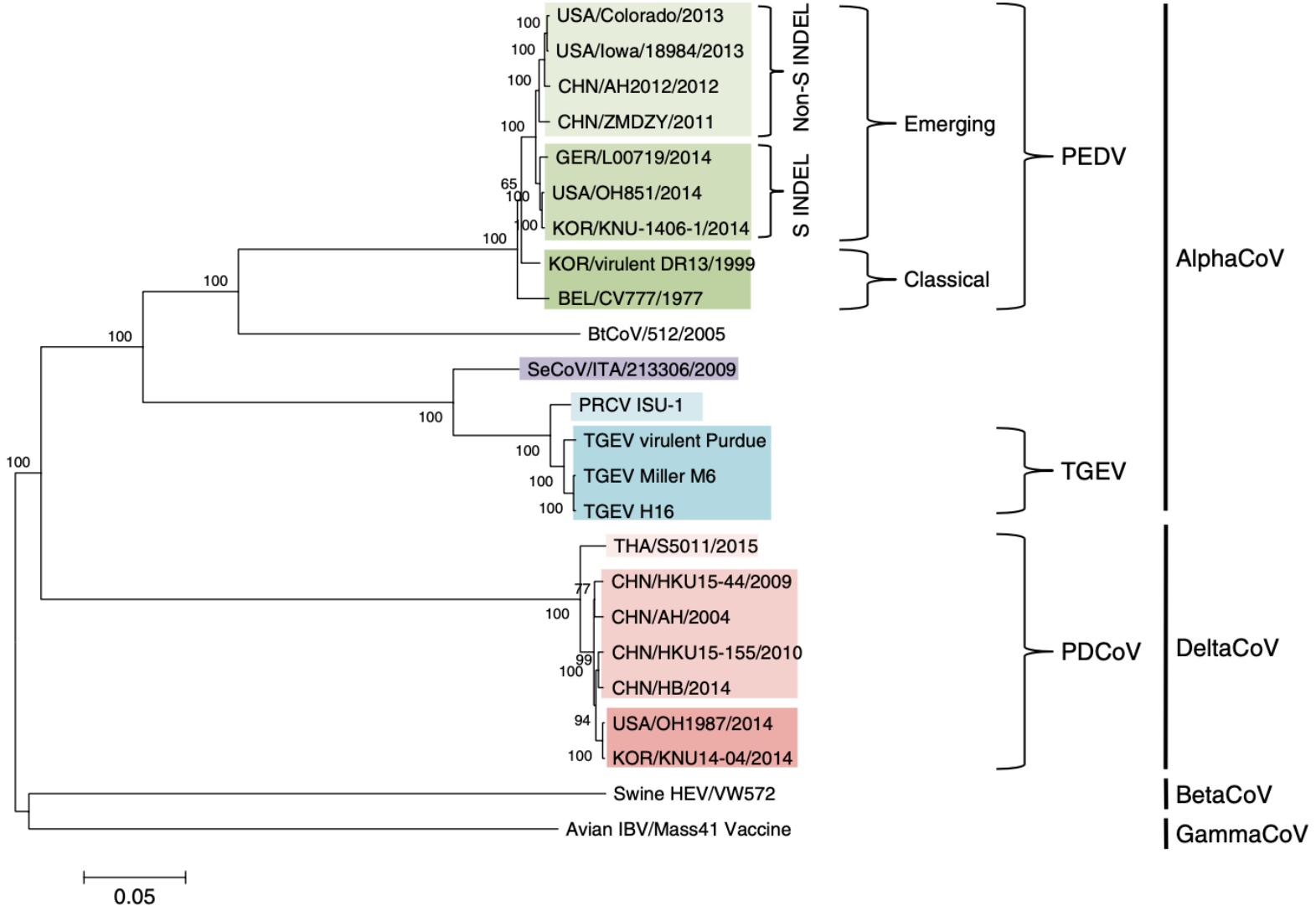
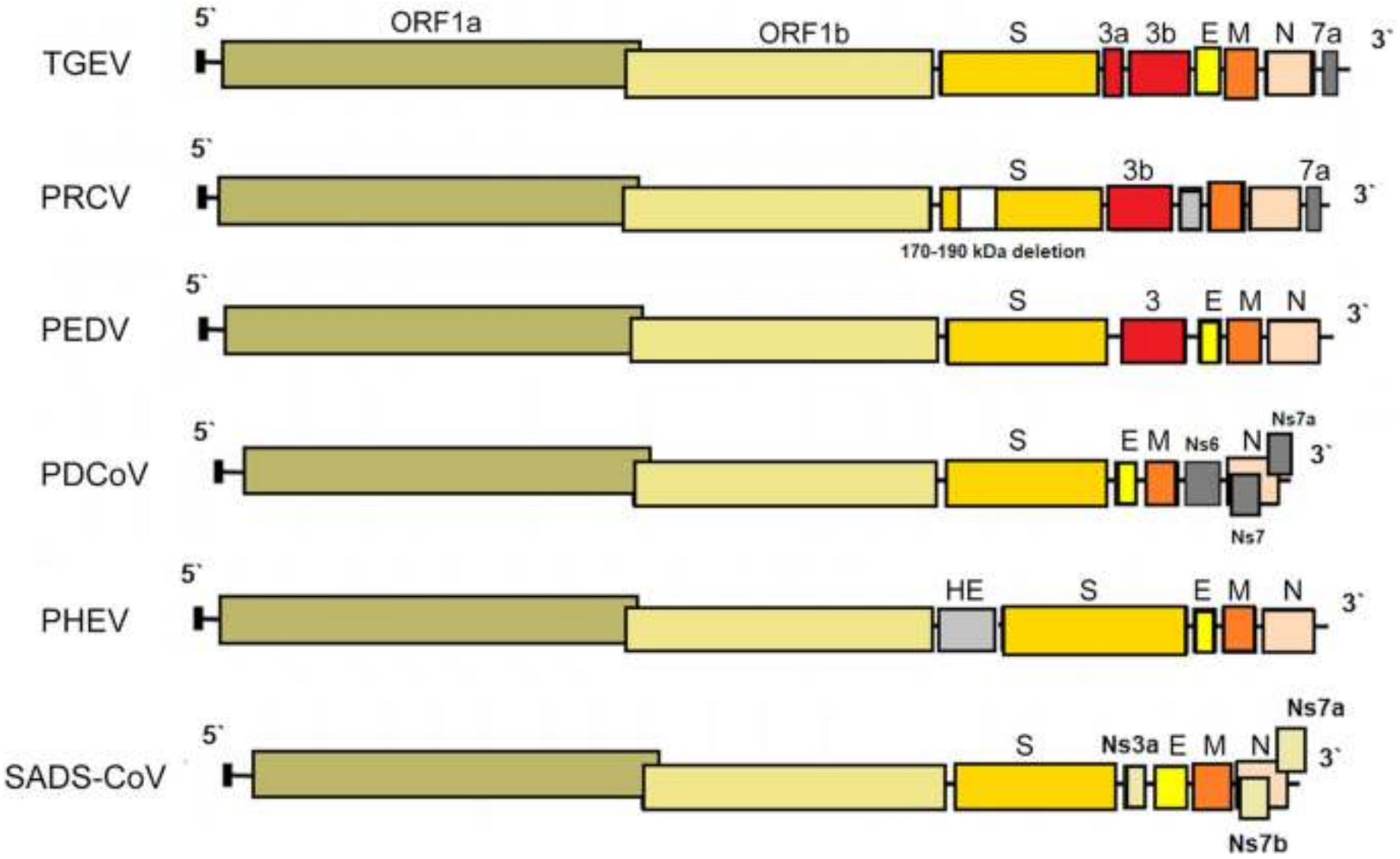


Figure 31.1 Phylogenetic tree of coronaviruses. The complete genomes of TGEV, PEDV, and PDCoV strains, closely related PRCV and a bat alphacoronavirus, a porcine HEV strain, and an avian IBV strain were selected. Multiple sequence alignments were performed using ClusterW, and a neighbor-joining phylogenetic tree supported with a bootstrap test of 1000 replicates was constructed using MEGA 6.0 software. The number on each branch indicates the bootstrap value. The scale represents the nucleotide substitutions per site. Color-shaded areas denote different swine virus clusters.



Turlewicz-Podbielska & Pomorska-Mól, 2021

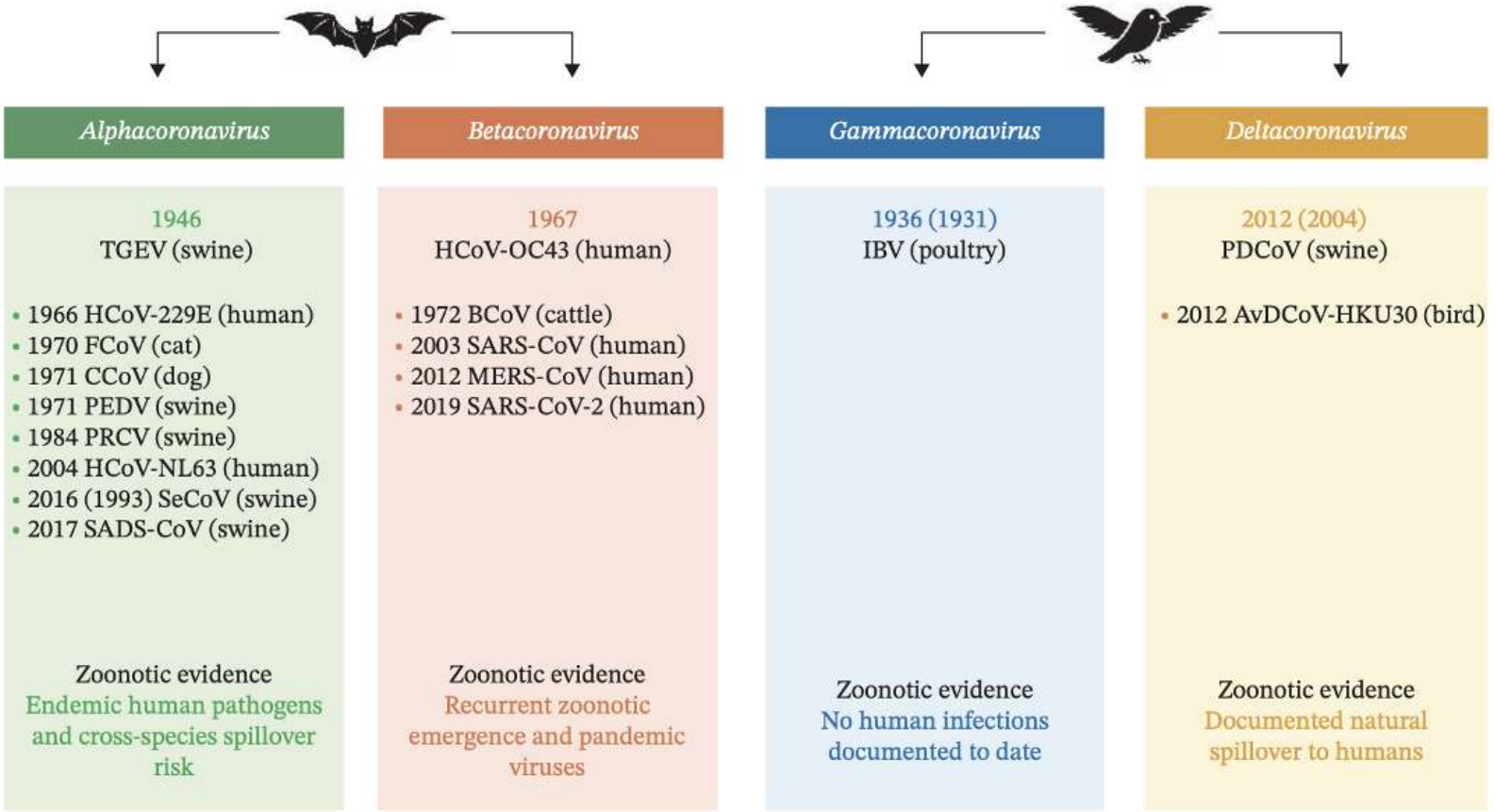


FIGURE 1 | Taxonomic classification of coronaviruses (CoVs). Phylogenetic evidence indicates that alpha- and betacoronaviruses originated primarily from bat reservoirs, whereas gamma- and deltacoronaviruses likely emerged from avian hosts, with subsequent host jumps to other animals or humans. Representative viruses, year of first description, main host species and zoonotic evidence are summarised. The zoonotic status for each genus is categorised based on documented empirical data, reflecting recurrent pandemic emergence (*Betacoronavirus*), natural spillovers or endemicity (*Alpha-* and *Deltacoronavirus*) and the current absence of human infections (*Gammacoronavirus*). Porcine enteric CoVs include porcine epidemic diarrhoea virus (PEDV), transmissible gastroenteritis virus/porcine respiratory coronavirus (TGEV/PRCV), swine enteric coronavirus (SeCoV) and swine acute diarrhoea syndrome coronavirus (SADS-CoV) within the *Alphacoronavirus* genus, as well as porcine deltacoronavirus (PDCoV) within the *Deltacoronavirus* genus. Other representative CoVs from different host species are shown, including human respiratory coronavirus: HCoV-229E, HCoV-OC43, HCoV-NL63, severe acute respiratory syndrome coronavirus (SARS-CoV), middle east respiratory syndrome coronavirus (MERS-CoV), severe acute respiratory syndrome coronavirus - 2 (SARS-CoV-2), infectious bronchitis virus (IBV), canine coronavirus (CCoV) and feline coronavirus (FCoV).

Transmissible Gastroenteritis (TGE) & Porcine Respiratory Coronavirus (PRCV)

縣府新聞
澄清專區
活動行事曆
縣府公告
招標公告
徵才公告
縣府電子公報
雲林良品

有關本縣豬隻罹患傳染性胃腸炎及豬流行性下痢案例，防疫所籲請依規定加強各項防疫措施。

發布單位：新聞處-公共關係科

雲林縣政府新聞參考資料103.1.8

邇來本縣部分豬場傳出仔豬隻感染傳染性胃腸炎及豬流行性下痢案例，其發病及死亡率均有明顯增加之情形，雲林縣動植物防疫所籲請轄內養豬業者加強牧場消毒及相關生物安全措施工作。

防疫所籲請縣內業者針對飼養場所必須確實執行下列生物安全措施，以防範疾病發生，包括：(一)、落實豬場消毒工作(二)、禁止各種運輸車輛、駕駛及不相關人員進入豬場(三)、禁止疫區人士進入畜牧場內；近期曾進入疫區之人員，回國後至少一週後始可進入飼養場所。(四)、勿購入來路不明之動物，以防病原引入。倘場內豬隻有異常病變或罹病率異常增加時，應即隔離並向所在地鄉鎮(市)動物防疫機關(公所)或雲林縣動植物防疫所通報，以為即時必要之防疫處置及避免疫情傳播。

https://www.yunlin.gov.tw/News_Content.aspx?n=1244&s=210321

Introduction

TGE

- Highly contagious, acute, **enteric viral disease**, characterized by vomiting, severe diarrhea, and high mortality in seronegative **piglets** less than 2 weeks of age.
- OIE listed disease

PRCV

- A natural mutant of TGE virus with **N-terminal deletion** on the **spike protein** resulting in **altered tissue tropism**, much less mortality, and widespread herd immunity against TGE.

Etiology

TGEV & PRCV

- Genus *alphacoronavirus*, subfamily of *Coronaviridae*
- Ss, positive-sense RNA genome of about 30 kb in length.

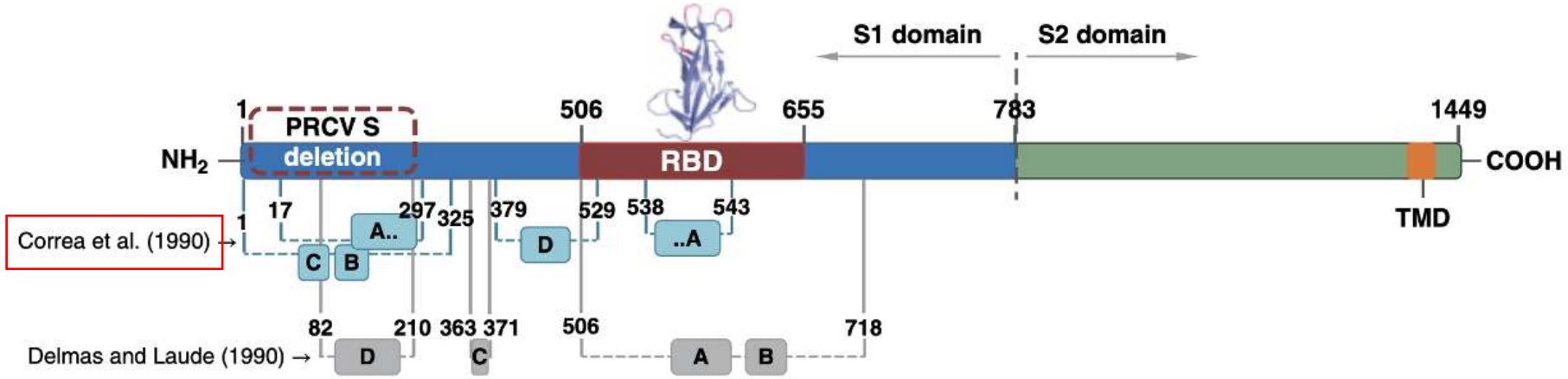


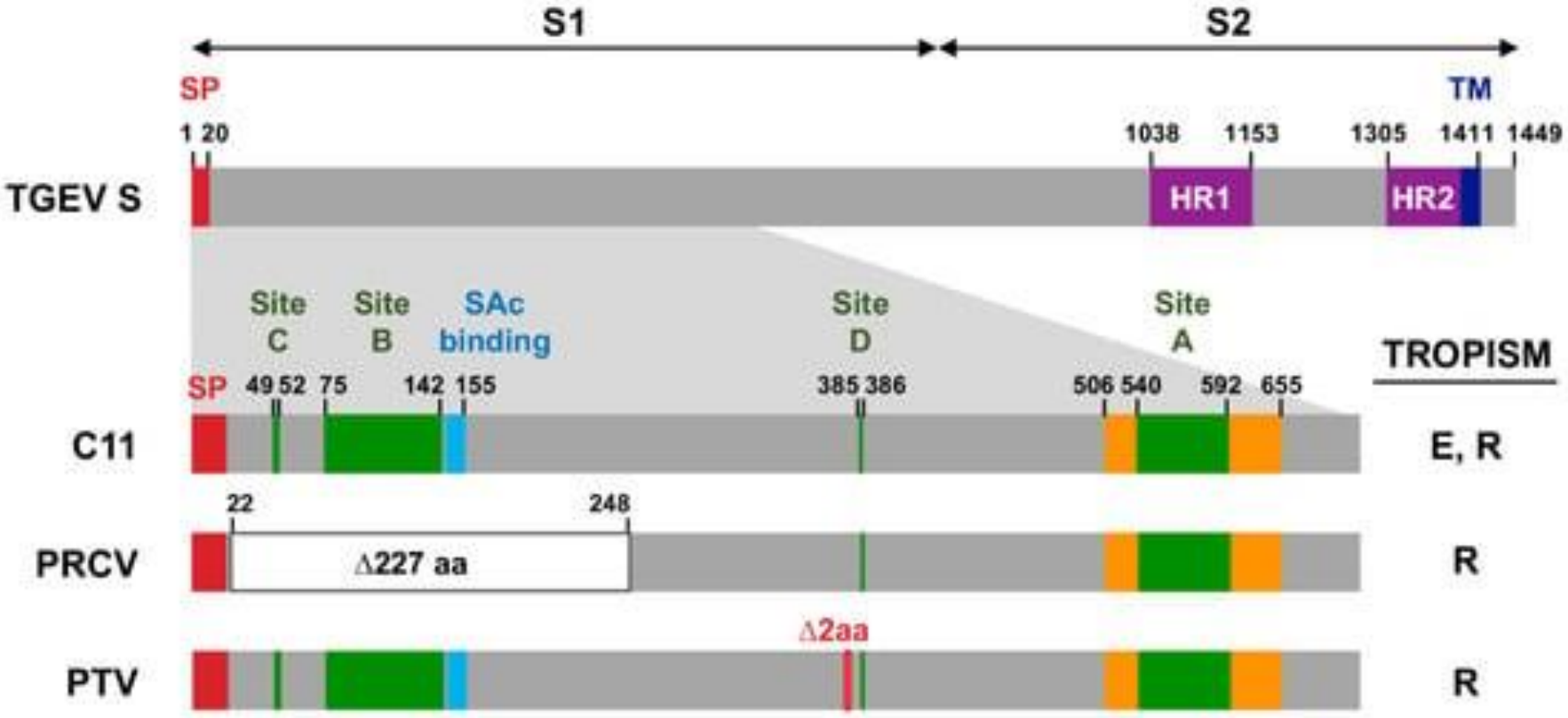
Figure 31.3 Schematic diagram of TGEV S protein. RBD, receptor-binding domain (shown in red); TMD, transmembrane domain (shown in orange). According to Correa et al.'s (1990) classification (light blue boxes), site A is discontinuous and located around positions 17 and 297 and 538 and 543, site B is located between amino acids 1 and 325, and site D is located between 379 and 529. The relative order of sites C and B and part of site A is not determined. According to Delmas and Laude's (1990) classification (light gray boxes), S protein antigenic sites D, C, and A/B are located between amino acids 82 and 210, between 363 and 371, and between 506 and 718, respectively. Dotted red box indicates a 227 aa deletion in the PRCV S protein. *Source:* Adapted from Correa et al. (1990) and Delmas and Laude (1990).

Diseases of Swine, 11th Edition

	TGEV	PRCV
Genome backbone	TGEV	TGEV
Deletion in spike	No	N-terminus (227 aa)
Loss activity of spike	No	~~Hemagglutination~~/ ~~Binding to sialic acid~~
Tissue tropism	GI tract	Respiratory tract
Cellular receptor	Aminopeptidase N (pAPN)	Aminopeptidase N (pAPN)

- Major neutralization epitopes (RBD domain and site A) on S protein are highly conserved for TGEV and PRCV.
- Only one serotype.

Tropism Alteration



Sanchez et al., 2019

Epidemiology – TGEV

- TGE were seen as early as 1943 but the first documented outbreaks in the US were reported in 1946.
- Characterized by **seasonal appearance during winter**, likely because TGEV is stable when frozen and more labile when exposed to heat or to sunlight.

Epidemiology – TGEV

- **Epidemic (流行性) TGE**: TGEV/PRCV **seronegative herd** → highly susceptible. Disease spreads rapidly and leads **to high mortality in young piglets**. Infected sows develop anorexia and **agalactia** which further contribute to piglet mortality.
- **Endemic (地方性) TGE**: **Persistence** of virus in a herd with continual or **frequent introduction** of susceptible pigs. Passive lactogenic immunity to suckling pigs leads to mild TGEV diarrhea and lower mortality (usually less than 10-20%). The source of virus remains unclear (carrier or reintro externally).

Epidemiology – PRCV

- PRCV was first isolated in Belgium in 1984; it then spread rapidly worldwide and persist in pig herds.
- Swine population density, distance between farms and season influence PRCV epidemiology.
- PRCV **persists** in closed breeding farms by regularly infecting newly weaned pigs, even in the presence of maternal antibodies
- Coincident **with the widespread dissemination of PRCV, the seroprevalence of TGEV in Europe has decreased**, to a low prevalence

Transmission-TGEV

- Swine that survive TGE develop antibodies against the virus but continue to excrete virus in their **feces (major)** or **nasal secretions** for at least 2-8 weeks. The longest record is 18 months
- Infected sows can transmit virus in their **milk** or **feces** to their piglets.
- Carriers of TGEV:
  - ✓ Dogs, cats and foxes can shed virus in their feces for variable periods.
  - ✓ TGEV was detected in the droppings of starlings for up to 32hours after feeding TGEV.
  - ✓ Feral pig population and insects

Pathogenesis – TGEV

- The virus enters via the **oronasal route**, is swallowed and reaches susceptible **epithelial cells** of the **small intestine**.
- Massive necrosis of jejunal enterocytes within 12-24h post-infection → reduction in enzymatic activity in the small intestine → acute **malabsorptive diarrhea**
- Resulting **villous atrophy** is dominant in jejunum and ileum but often absent in the proximal duodenum.
- Extraintestinal sites for TGEV replication include lungs (alveolar macrophages) and mammary tissues.

Pathogenesis – TGEV

- All age groups are affected but clinical severity is **age-dependent.**
- In piglets: malabsorptive diarrhea → severe & fatal dehydration + metabolic acidosis & hyperkalemia with consequent cardiac dysfunction.
- Slower replacement of the destroyed villi by cells from crypts in newborn pigs also accounts for the age-dependent susceptibility.

Pathogenesis – PRCV

- PRCV has a **tropism for the respiratory tract**.
- It replicates to high titers in porcine lungs (1×10^{7}–10^{8} $TCID_{50}$) in type 1 and 2 pneumocytes.
- PRCV also infects epithelial cells of the nares, trachea, bronchi, bronchioles, alveoli, and, occasionally, alveolar macrophages.
- PRCV exhibits **limited intestinal replication** and spreading.

Clinical Presentation – TGEV

Epidemic (流行性) TGE

- **Short incubation** period: 18 hours to 3 days
- Vomiting and profuse watery, yellowish diarrhea, with rapid loss of weight, dehydration, and high morbidity and mortality in pigs under 2 weeks of age.
- Clinical severity:
  - ✓ < 7 days of age: die within 2–7 days
  - ✓ > 3 weeks of age: survive but may remain stunted
  - ✓ Finishing swine & sow: inappetence, transient diarrhea & vomiting

Clinical Presentation – TGEV

Endemic (地方性) TGE

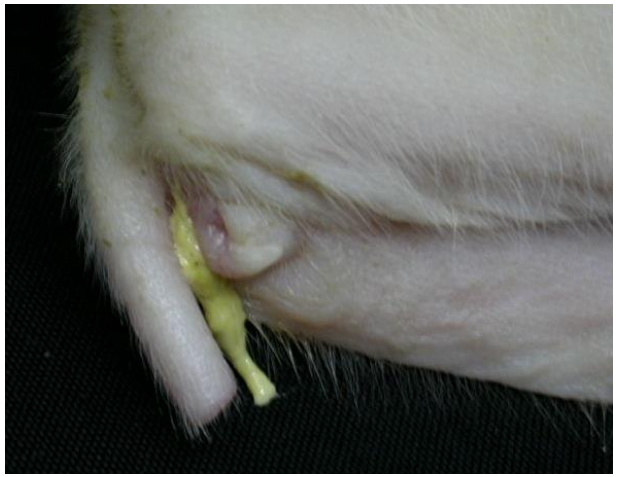
- Less severe clinical disease than those in seronegative pigs of the same age.
- Mortality is low.

Clinical Presentation – PRCV

- Mostly **subclinical** and self-limiting respiratory infection.
- Clinical signs:
  - ✓Coughing, abdominal breathing, dyspnea
  - ✓Depression +/- anorexia
  - ✓Slightly decreased growth rates
- The clinical severity are influenced by coinfection with other pathogens.

Lesions – TGEV

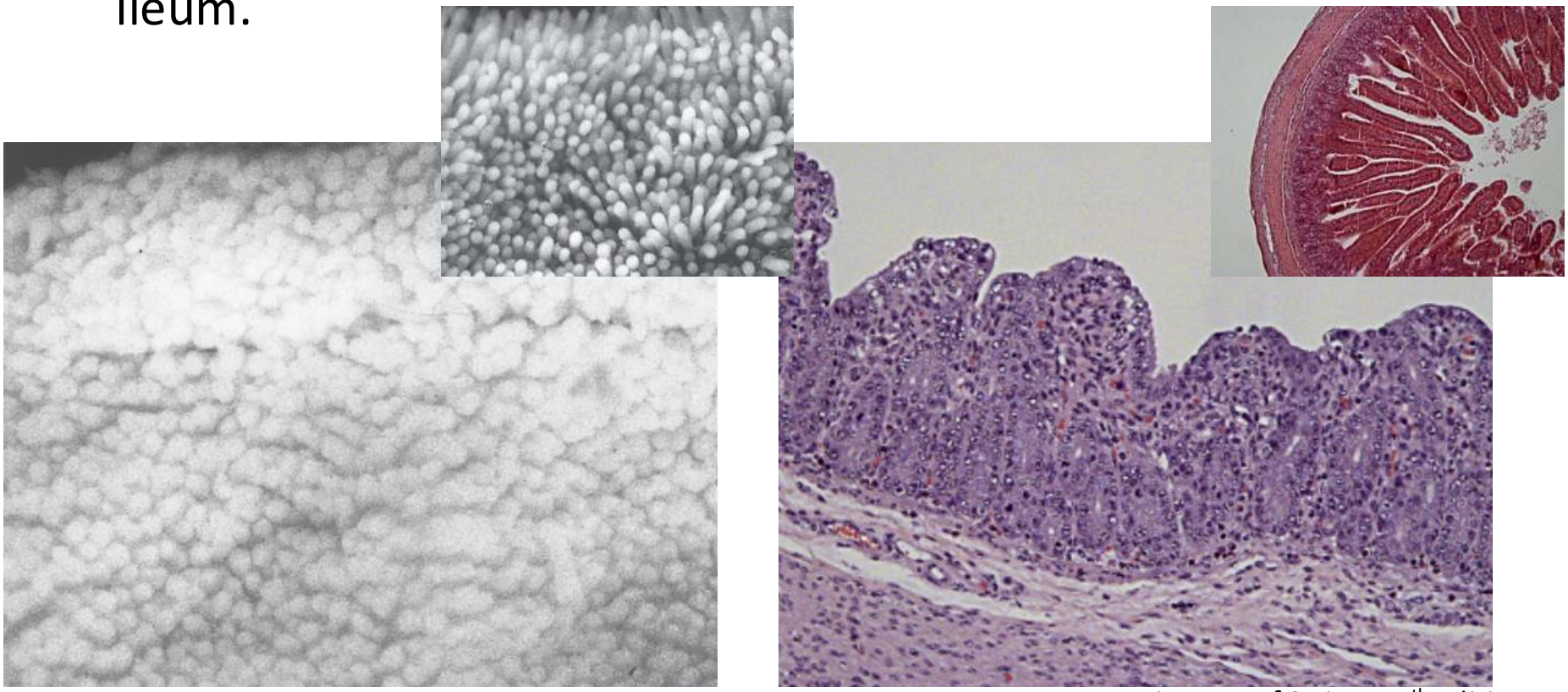
- Gross lesions are confined to the GI tract:
  - ✓The stomach may be empty because of vomiting but sometimes is filled with curdled milk.
  - ✓The intestinal wall is very thin and nearly transparent.
  - ✓Chyle seldom is present in mesenteric lymphatics since fat absorption is impaired.





Lesions – TGEV

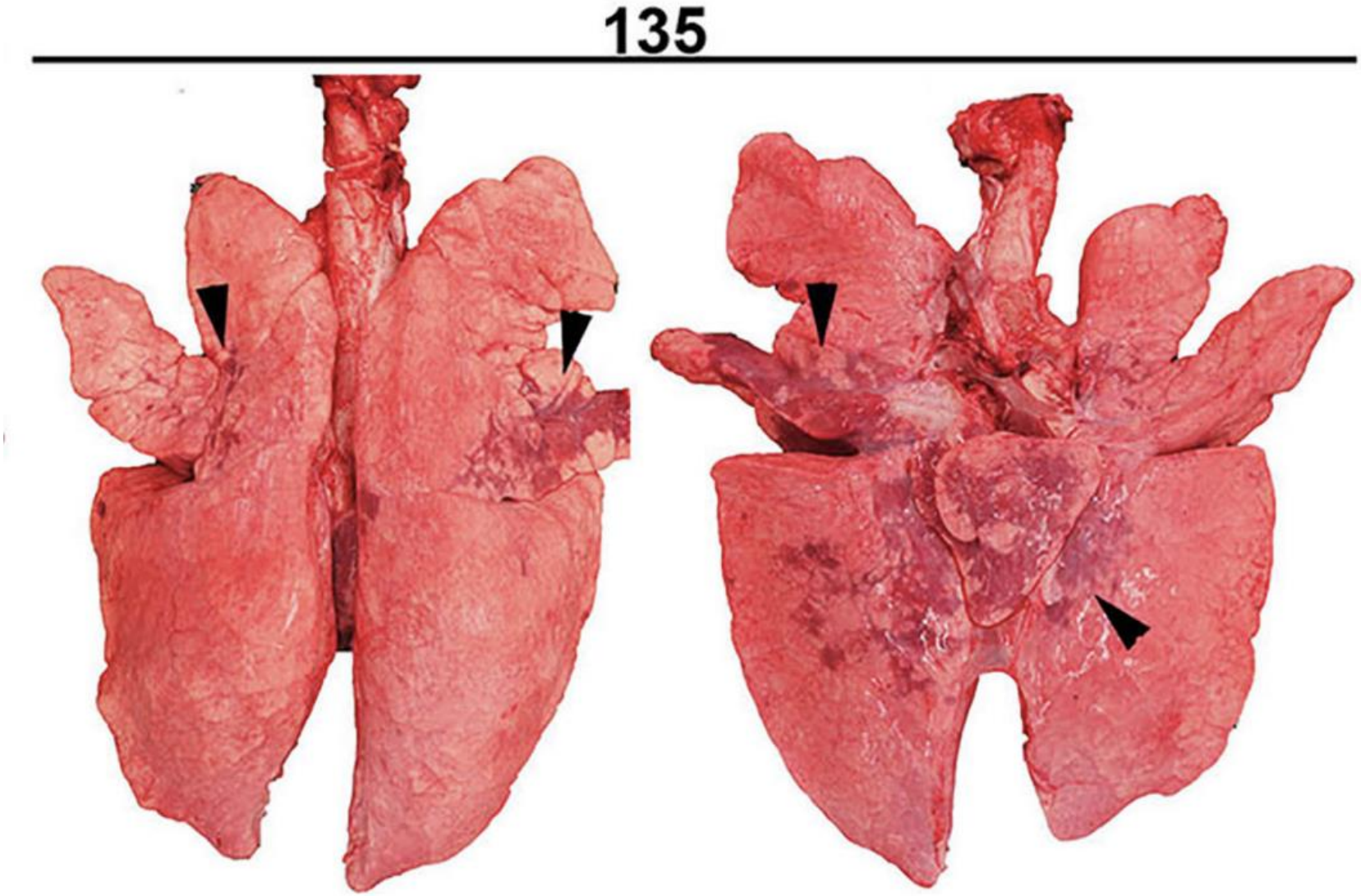
- Microscopic alterations include a **marked villous atrophy**, perhaps with squamous metaplasia of epithelium, and elongation of crypts in scattered areas of the jejunum and ileum.



Diseases of Swine, 11th Edition

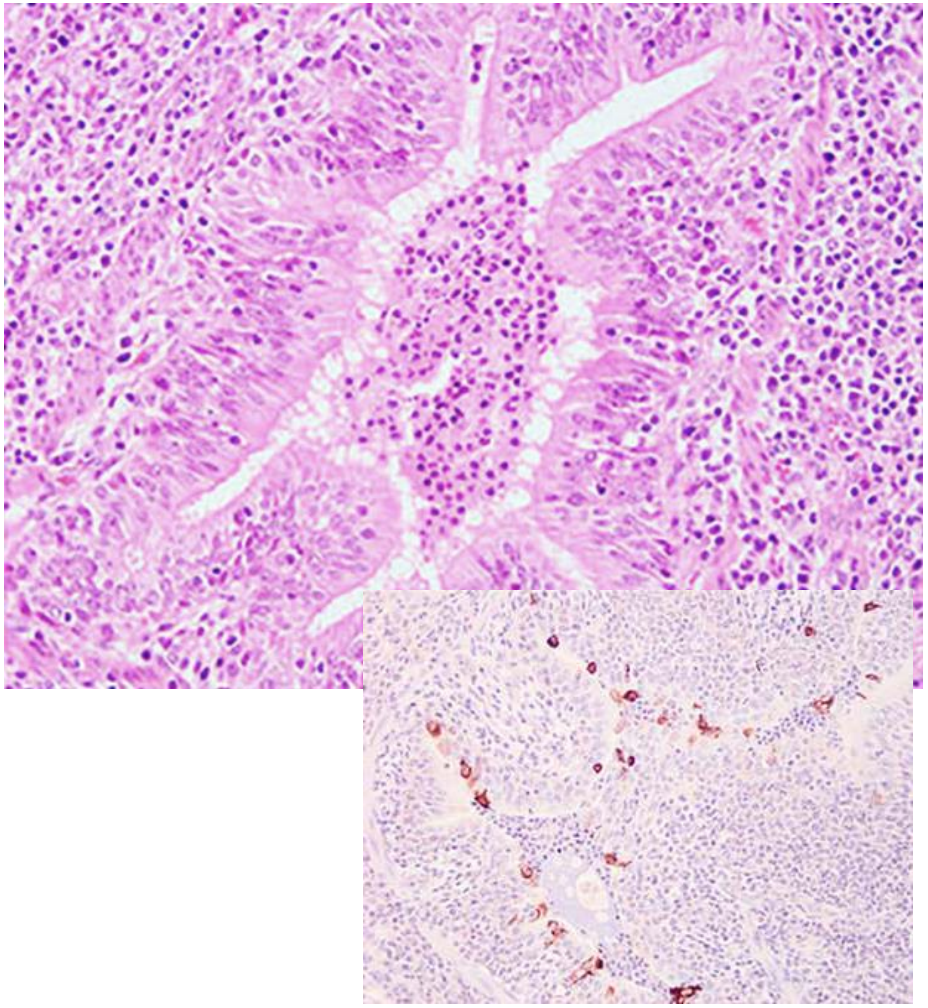
Lesions – PRCV

- The PRCV-induced lesions are generally limited to the lungs.
- Gross: consolidation of the affected lung lobes
- Micro:
  - ✓Thickening of the alveolar septa by inflammatory infiltration
  - ✓Type 2 pneumocyte hypertrophy and hyperplasia
  - ✓Accumulation of necrotic cells and inflammatory leucocytes in alveolar and bronchiolar lumina due to airway epithelial necrosis
  - ✓Peribronchiolar or perivascular lymphohistiocytic cuffing

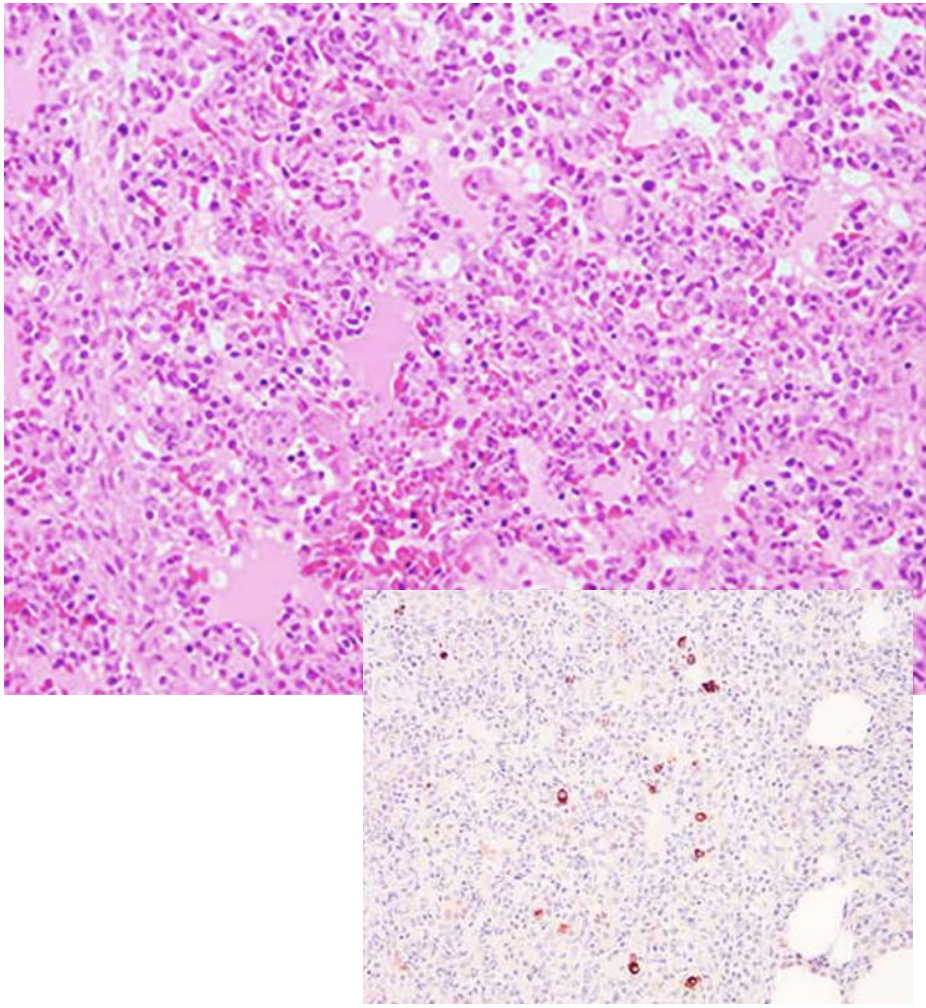


Keep et al., 2022

Alveola



Bronchiole



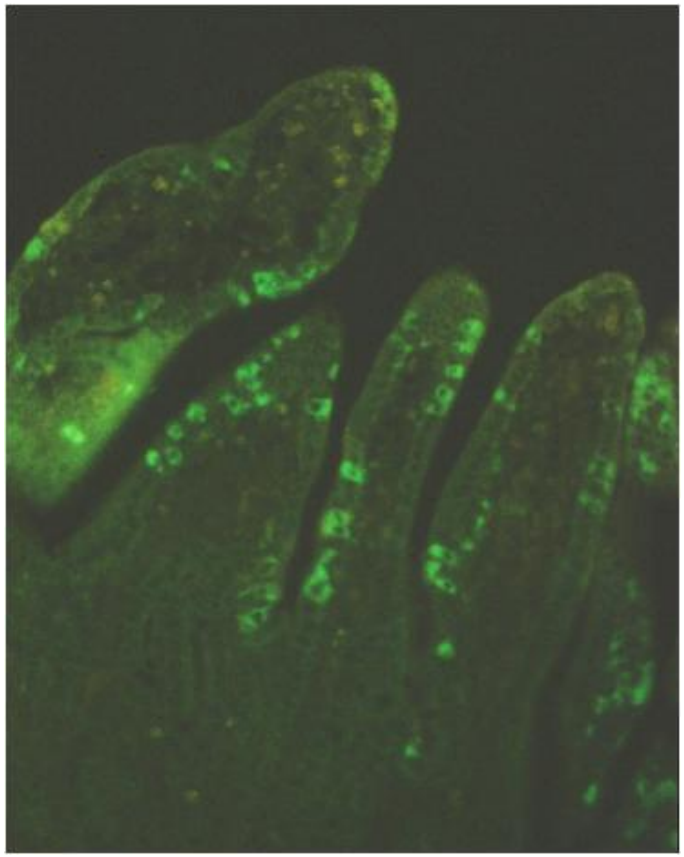
Keep et al., 2022

Laboratory Diagnosis

- Virus detection (feces or lesions for TGEV & nasal swab or lung homogenate for PRCV)
  - ✓ Identification of infectious) virion: IEM, virus isolation

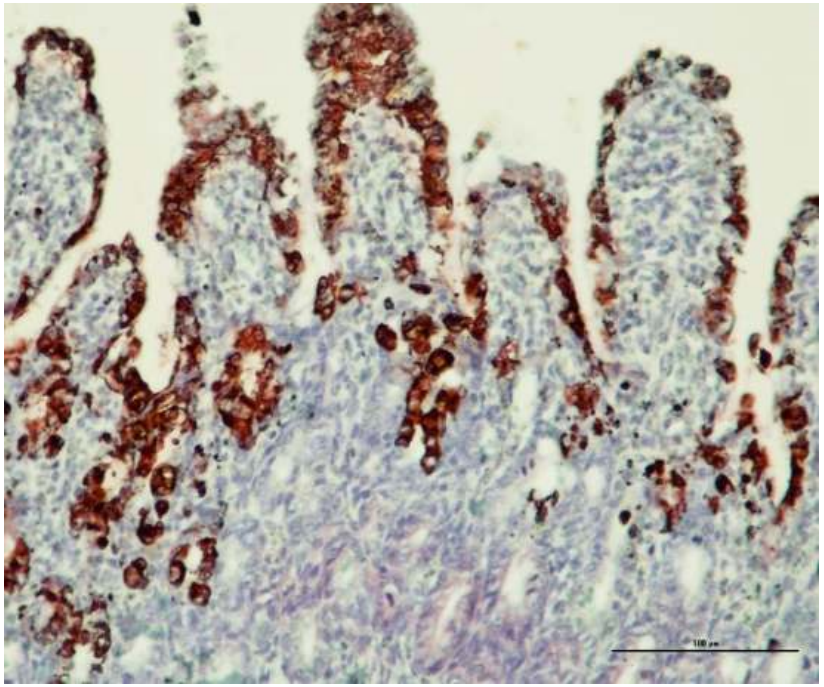
    *using PK15, ST, or porcine thyroid cells; blind passages
  - ✓Identification of antigen: ELISA, IF, IHC
  - ✓Identification of nucleic acids: RT-PCR, RT-qPCR
- Serological test
  - ✓ELISA: blocking ELISA for differentiation TGEV vs PRCV
  - ✓Virus neutralization

Immunofluorescence stain



MAb against TGEV

Immunohistochemical stain



MAb against TGEV

Immunity

- Active immunity to TGEV induced by infection: secretory **IgA** (sIgA) by gut plasma cells, which may last for at least 6 months to years.
- **PRCV-induced active immunity to TGEV**: prior infection of nursing or weaned pigs with PRCV provides partial immunity against TGEV challenges due to **migration of PRCV IgG and IgA antibody-secreting cells** from bronchus-associated lymphoid tissue **(BALT) to the gut**.
- Passive immunity to TGEV: recovered swine transmit passive immunity to their suckling pigs via colostrum or milk on the basis of **gut-mammary immnunologic axis**.

Prevention & Control

- There is no specific treatment for TGEV.
- Strict biosecurity to prevent introduction of carriers or addition of infected population.
- After onset of TGE
  - ✓Animals due to farrow in more than 2 weeks: purposely expose to virulent virus
  - ✓Animals due to farrow less than 2 weeks: provide facilities and management procedures that will avoid exposure to TGEV until at least 3 weeks post-farrowing

Prevention & Control

- In endemic area

Both live-attenuated or inactive TGEV vaccines are available.

✓Vaccinated pregnant seropositive sows in gestation

✓Alterations in management to break cycle of infection by eliminating reservoirs of susceptible pigs in a unit.

Eg. 1. Whole-herd exposure to virus and allowing no introductions into the herd for a 60-day period.

2. Depopulation during a hot month accompanied by thorough cleaning and disinfection. Then, after allowing the facilities to be free of swine for several weeks, restocking with serologically negative stock.

有關豬隻傳染性胃腸炎之敘述，下列何者錯誤？

(A) 由betacoronavirus感染所致
(B) 哺乳期仔豬死亡率高
(C) 可見小腸絨毛萎縮
(D) 豬隻是唯一可自然感染本病的動物

下列豬冠狀病毒中，何者與新冠病毒（COVID-19）親緣最為相近？

(A) 豬流行性下痢病毒
(B) 豬傳染性胃腸炎病毒
(C) 血球凝集腦脊髓炎病毒
(D) 豬戴爾他冠狀病毒（porcine deltacoronavirus）

有關豬傳染性胃腸炎的病灶，何者正確？①小腸黏膜上皮細胞壞死 ②常伴隨小腸腺窩上皮細胞壞死 ③小腸絨毛萎縮及融合

(A) 僅①②
(B) 僅①③
(C) 僅②③
(D) ①②③

下列何種豬病不是冠狀病毒所引起？

(A) 傳染性胃腸炎
(B) 豬水疱病
(C) 血球凝集腦脊髓炎
(D) 豬流行性下痢

下列有關豬呼吸道冠狀病毒(porcine respiratory coronavirus)之敘述,何者錯誤?

(A) 可感染所有年齡豬隻
(B) 可感染呼吸道及消化道
(C) 多為不顯性感染
(D) 豬隻感染後可產生中和傳染性胃腸炎(transmissible gastrenteritis)病毒之中和抗體

下列有關豬呼吸道冠狀病毒(porcine respiratory coronavirus)之敘述,何者錯誤?

(A) 可感染所有年齡豬隻
(B) 可感染呼吸道及消化道
(C) 多為不顯性感染
(D) 豬隻感染後可產生中和傳染性胃腸炎(transmissible gastrenteritis)病毒之中和抗體

以免疫螢光法診斷豬傳染性胃腸炎時，應採下列何部位檢體最為適當？

(A) 十二指腸
(B) 空腸
(C) 結腸
(D) 胃

豬流行性下痢與豬傳染性胃腸炎的臨床症狀幾乎無法區別，而且二者的致病原都是下列何種病毒？

(A) coronavirus
(B) paramyxovirus
(C) picornavirus
(D) rhabdovirus

下列有關豬傳染性胃腸炎之致病機序，何者錯誤？

(A) 以食入感染為主
(B) 致病原可感染胃、小腸及大腸之黏膜上皮
(C) 豬隻下痢之主因為腸道絨毛萎縮
(D) 高致病性病原對消化道內蛋白質分解酶具有較強之抵抗力

下列有關豬呼吸道冠狀病毒（porcine respiratory coronavirus，PRCV）與傳染性胃腸炎病毒（transmissible gastroenteritis virus，TGEV）的敘述，何者錯誤？

(A) 兩者的基因非常相似
(B) PRCV是TGEV的變異株
(C) 兩者均造成小腸絨毛萎縮
(D) 兩者間具有交叉反應

Porcine Epidemic Diarrhea (PED)

小豬仔大量下痢病死 端午前後豬隻供應將短缺

上下游記者楊鎮宇 · 漁業.畜牧 · 2014 年 02 月 12 日

最近寒流來襲，不但家長需注意小孩感染腸病毒，連養豬業者也擔心小豬仔被稱為「仔豬殺手」的流行性下痢（PED）傳染。從去年底至今，全台灣有207個養豬場傳出流行性下痢疫情、34800多頭仔豬死亡，養豬場損失慘重，端午、中元前後的豬價可能也將起波動。

小豬大量死亡 疫情猶如「秋風掃落葉」

這波仔豬得流行性下痢的疫情，從南部開始，雲林、屏東、台南等地較嚴重，南投、彰化、苗栗等地也有零星災情。光是雲林縣，從去年十一月傳出疫情至今，就有超過一萬七千多頭仔豬死亡。中華民國養豬協會常務監事潘連周說，屏東大小豬場幾乎都受感染，仔豬只要一得病，八成無法倖存，「三十年來第一次碰到這麼嚴重的流行性下痢疫情」。

流行性下痢（porcine epidemic diarrhea；PED）是由冠狀病毒引起的疾病，得病的豬隻會有嘔吐、下痢的情況，仔豬若得病會大量脫水，死亡率非常高。

最新文章



神農夫婦打造全台最大有機農場，銀川米340公頃有機田，讓農民與每粒米都有歸屬

https://www.newsmarket.com.tw/blog/46549/

Introduction

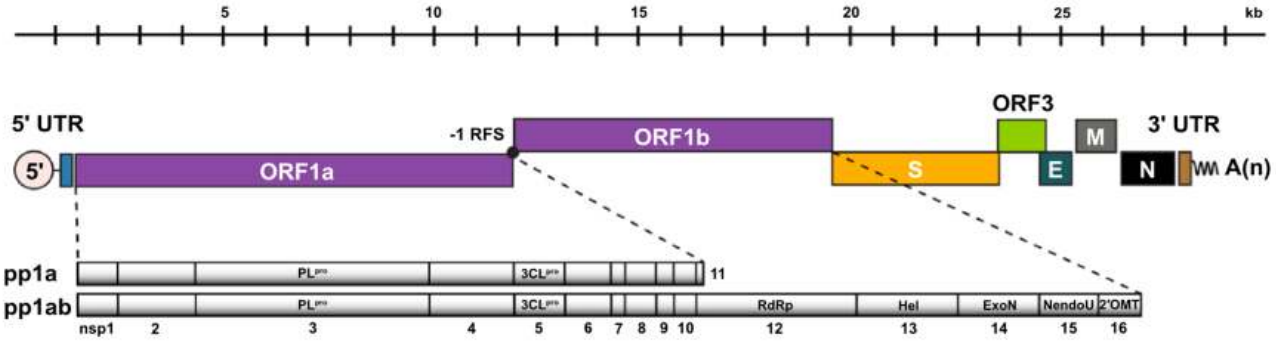
- Highly contagious, acute, **enteric viral disease**, characterized by vomiting, severe diarrhea, and high mortality in seronegative **young piglets**.
- First described in 1971 in the United Kingdom—prototype CV777. Historically, no economic importance.
- The emergence of highly virulent non-S INDEL PEDV strains in China in late 2010 raised its significance in swine industry.

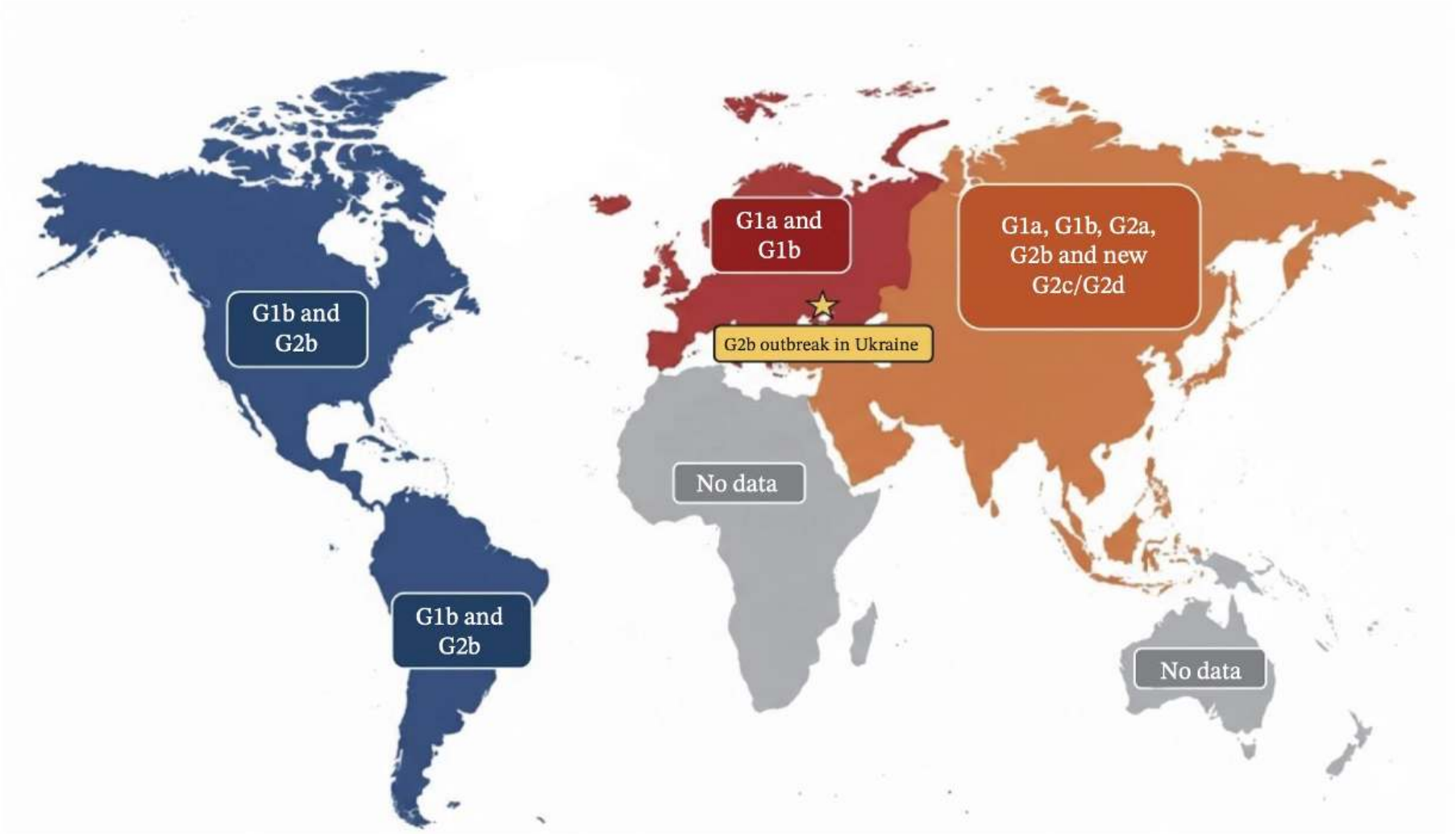
Etiology

Porcine epidemic diarrhea virus (PEDV)

- Genus *alphacoronavirus*, subfamily of *Coronavirinae*
- Ss, positive-sense RNA genome of about 30 kb in length.
- Phylogenetic analysis based on the S gene:
  - ✓G1a/b: Historic PEDV (e.g. CV777) & S-INDEL
  - ✓G2a/b/c/d: Emerging highly pathogenic

  ** Immune pressure from widespread vaccination*
- One serotype




G1b and G2b
G1b and G2b
G1a and G1b
G2b outbreak in Ukraine
G1a, G1b, G2a, G2b and new G2c/G2d
No data
No data

Pathogenesis

The pathogenesis of PED is related to age, virus strains, inoculation route and doses.

Table 31.1 Comparative pathogenesis of different clusters of PEDV in experimentally infected piglets (younger than 6 days of age).

PEDV strain	Inoculum/dose per pig	Pig type/age (day) at inoculation	Villous atrophy (VH:CD ratios)	Onset of clinical signs (hpi)	Vertical location of PEDV		Longitudinal distribution of PEDV		References
					Villous	Crypt	D, J, I	C	
Classical									
CV777	Fecal suspension/4 $\log_{10}$ PID	CDCD/2–3	Moderate to severe (1.5–4.2)	22–36	+++ (entire)	+	D, J, I (cont)	+	Coussement et al. (1982)
SNUVR971496	Cell culture (P3)/6.8 $\log_{10}$ $TCID_{50}$	Colostrum-deprived/1	Severe (1.1–3.3)	12–36	+++ (entire)	–	D, J, I (cont)	–	Kim and Chae (2003)
Non-S INDEL									
IN19338	Cell culture (P7)/0.056–5600 $TCID_{50}$	Conventional/5	Severe (1.2–1.7)	24	+++ (entire)	NR	D, J, I (cont)	NR	Thomas et al. (2015)
PC22A	Cell culture (P3)/1–4 $\log_{10}$ PFU	CDCD, conventional/3–4	Severe (0.8–2.3)	< 24	+++ (entire)	+	D, J, I (cont)	+	Liu et al. (2015b)
S INDEL									
Iowa106	Fecal suspension/10–12 $\log_{10}$ GE	Conventional/4	Moderate to severe (1.4–5.4)	24–72	++ (entire)	–	D (patchy), J, I (cont)	–	Lin et al. (2015)
IL20697	Cell culture/5 $\log_{10}$ $TCID_{50}$	Conventional with milk replacer/5	Mild	NR	+/++	NR	D, J (NR), I (patchy)	+	Chen et al. (2016a)

Source: This table is adapted/updated from Saif (1989). Reproduced with permission of Taylor and Francis.
PID, pig infectious dose; $TCID_{50}$, 50% tissue culture infectious dose; PFU, plaque-forming unit; GE, genomic equivalent; CDCD, Cesarean-derived colostrum-deprived; SPF, specific pathogen free; VH: CD, villous height/crypt depth ratio; hpi, hours' post inoculation; D, duodenum; J, jejunum; I, ileum; cont, continuous; NR, not reported; –, +, ++, and +++ denotes none, less than 30%, 30–60%, and more than 60% of villous enterocytes that were positive for PEDV antigens, respectively.

Pathogenesis

- PEDV infection via **oral-fecal route** results in loss of **enterocytes in the small intestine**, leading to malabsorptive **diarrhea**.
- All age groups can be affected. Factors contribute to the **higher susceptibility of neonatal piglets** than weaned pigs:
  - ✓ **Slower turnover** of villous enterocytes in neonatal piglets (5–7 days) compared with 2–3 days in 3-week- old weaned pigs
  - ✓ Increased numbers of intestinal stem cells and proliferation of crypt cells occurred later (3 days post- PEDV) in neonatal piglets than in weaned pigs (1 day post-PEDV)
  - ✓ Deficiency in innate immunity in suckling pigs compared with weaned pigs.

Clinical Presentation

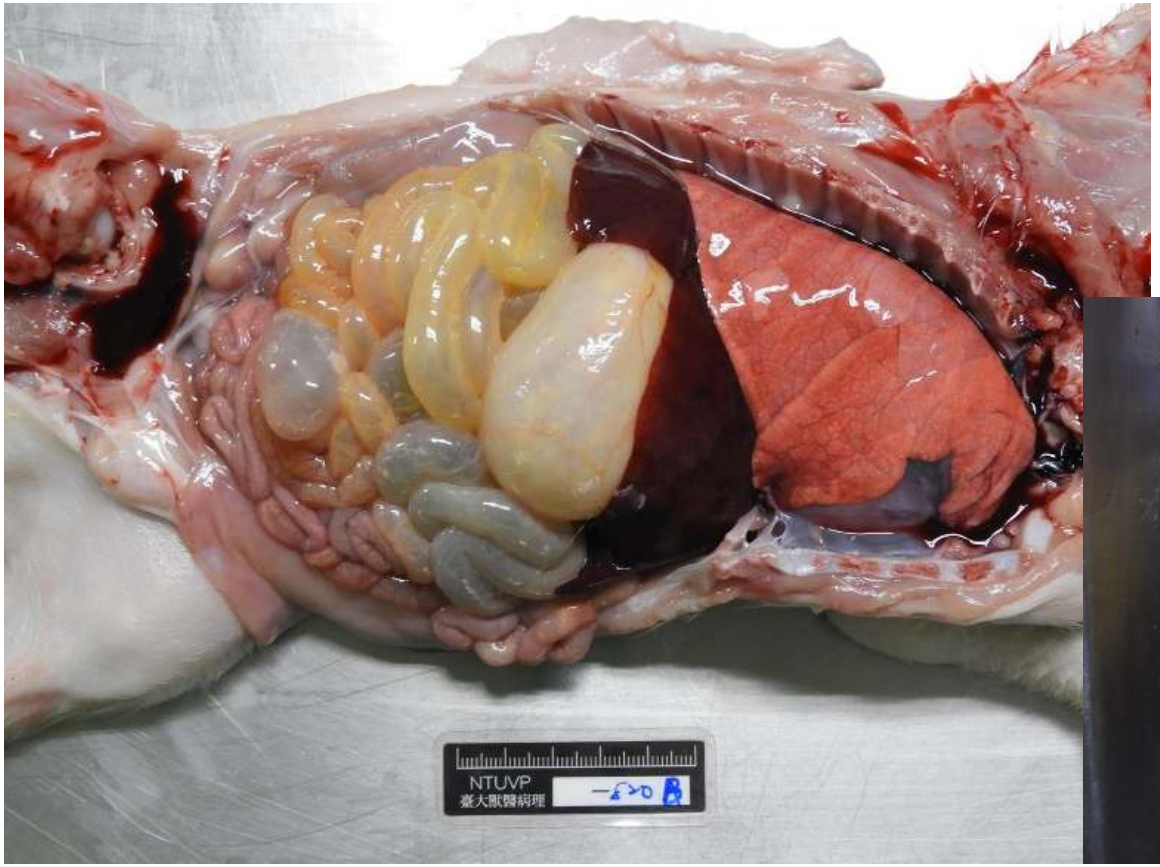
Similar to those seen in TGE, including **watery diarrhea**, **vomiting**, anorexia and depression.



https://www.youtube.com/watch?v=Yp9Bxf0F57Q

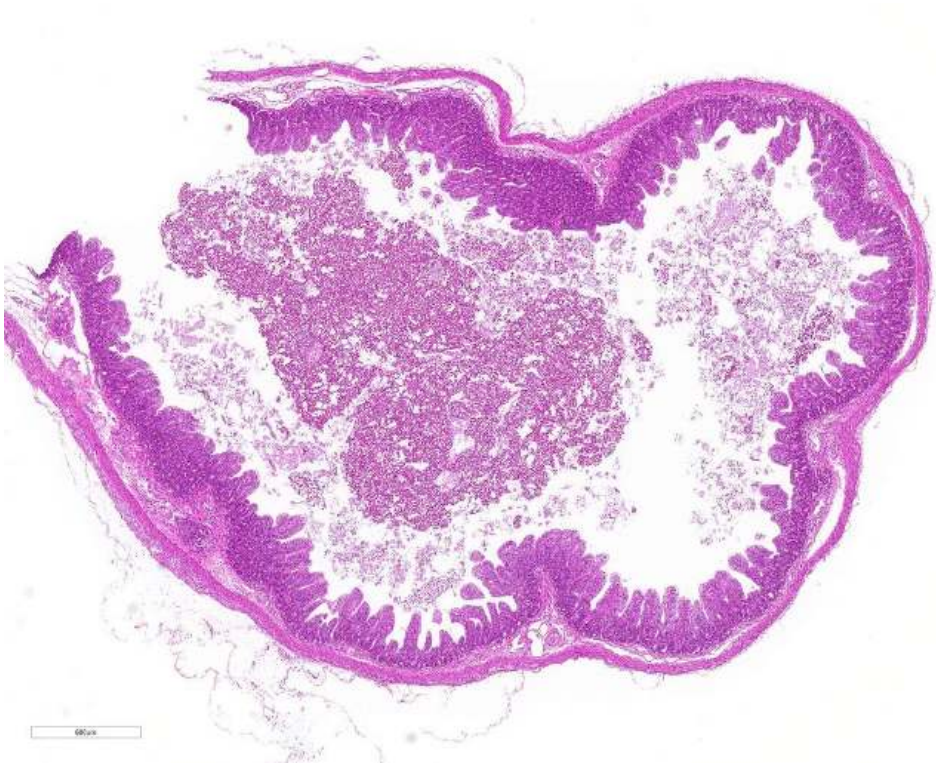
Lesions

Gross lesions are confined to the **small intestine** that is distended with watery, yellowish fluid.

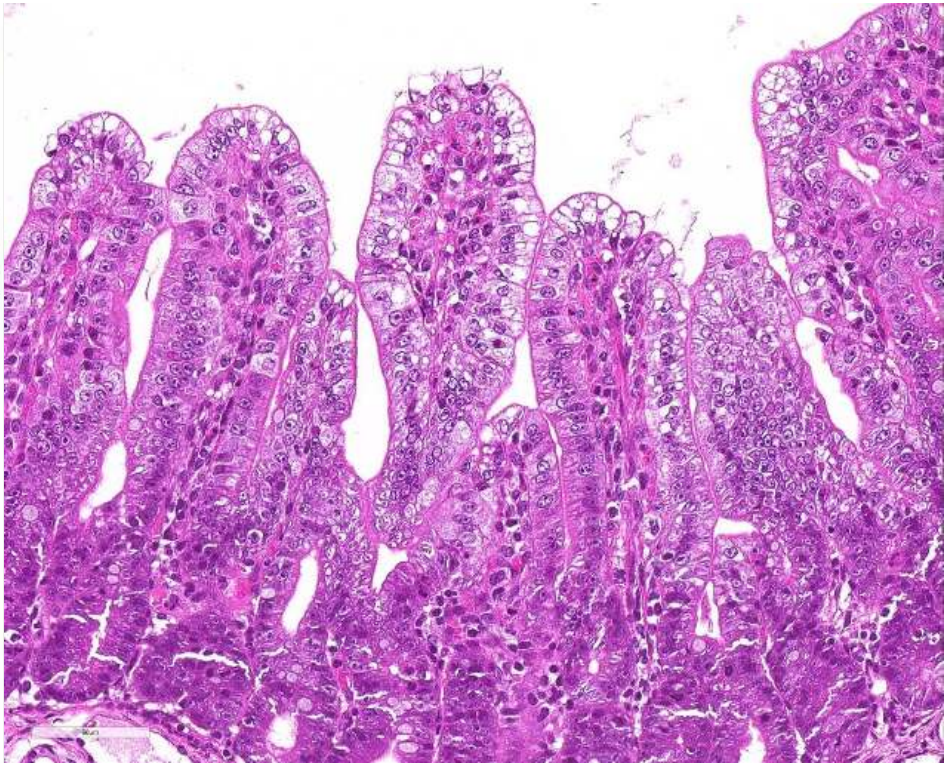




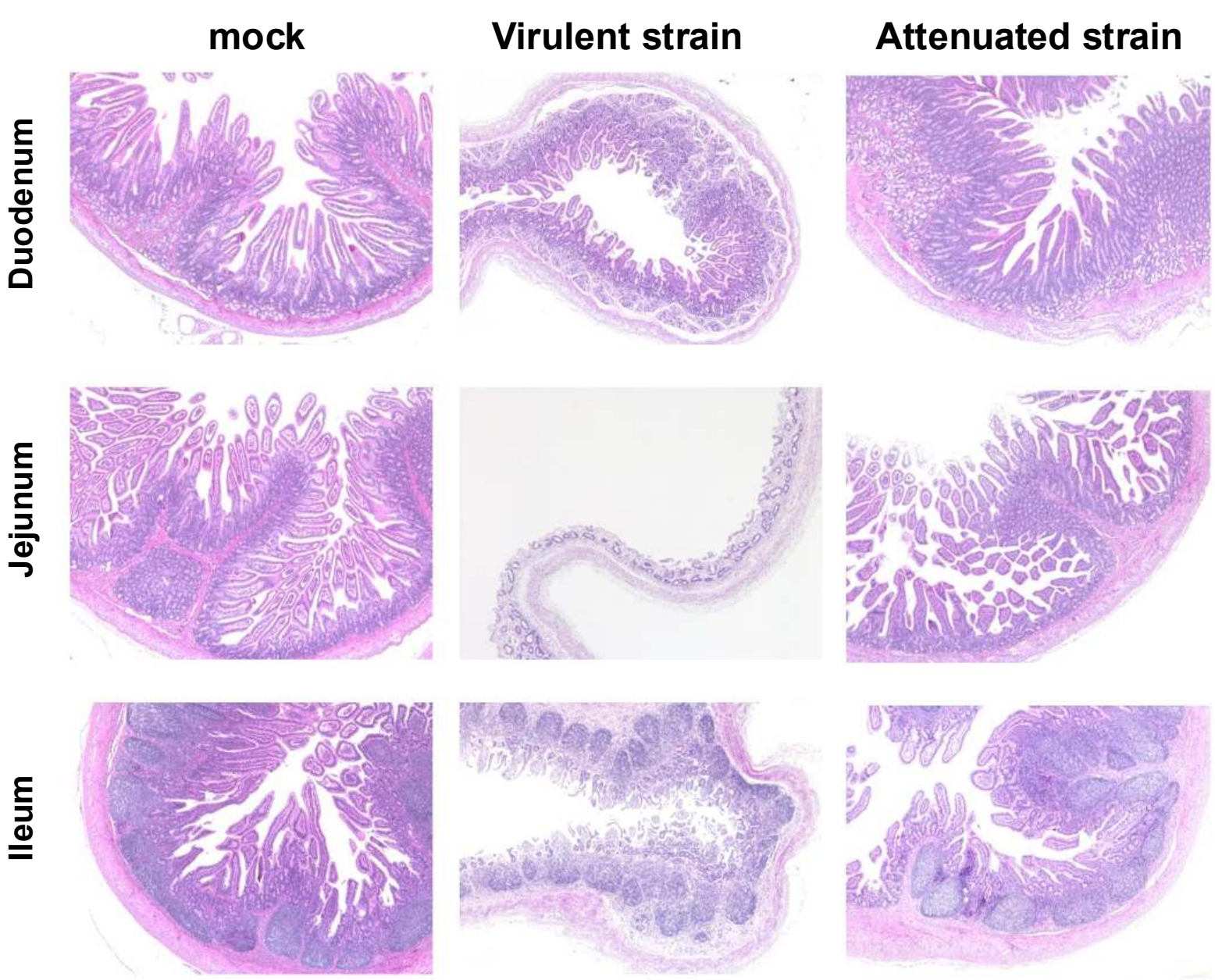
Lesions



Low magnification of a cross section of jejunum shows marked **villar blunting** and fusion, as well as hemorrhage within the lumen.

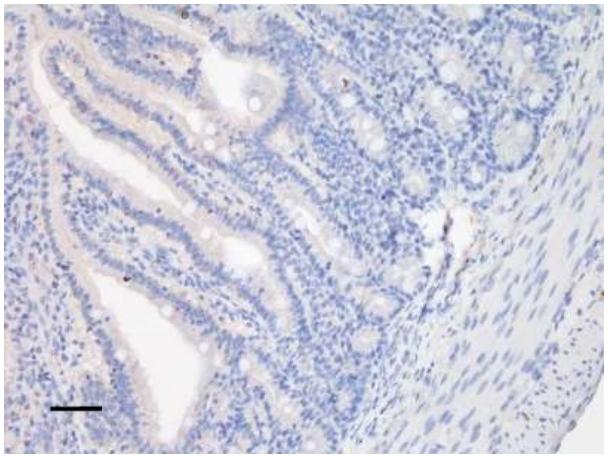


There is **degeneration of villar tip enterocytes** characterized by the presence of large vacuoles.

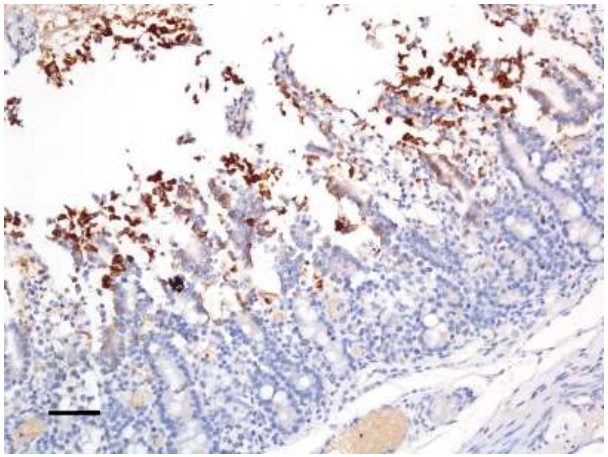


The histopathology is very similar to that observed in TGE and PDCoV

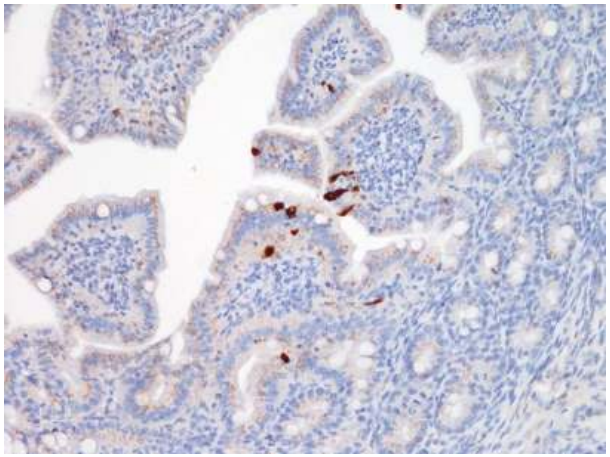
mock



Virulent strain



Attenuated strain



PEDV antigens were detected by using mAB against N proteins.

Laboratory Diagnosis

- Virus detection (feces or lesions)
  - ✓ Identification of infectious) virion: IEM, virus isolation

    *Vero cells with trypsin treatment; blind passages
  - ✓Identification of antigen: ELISA, IF, IHC
  - ✓Identification of nucleic acids: RT-PCR, RT-qPCR

- Serological test (oral, fecal, blood)
  - ✓ELISA
  - ✓Virus neutralization

Immunity

- Although PED occurs in pigs of all ages, piglets up to 1 week of age may experience high mortality and need to be protected by maternal antibodies, especially VN and **sIgA**, via colostrum and milk from immunized dams.
- **Lactogenic immunity** is induced in sows by intestinal infection with PEDV, which then activates the **gut– mammary–sIgA** axis.
- Classical PEDV vaccines were not efficacious in protecting pigs against the highly virulent non-S INDEL PEDV infection.

Prevention & Control



Zoetis

- Vaccines
- Pregnant sows more than two weeks away from parturition can be **intentionally exposed in the face of an outbreak** (feedback of watery feces or intestinal content from acutely affected pigs). However, this procedure has the obvious **risk of spreading other diseases**.
- Sanitary and quarantine measures may slow the spread of PED.
- There is no effective treatment other than good care and the provision of adequate water to combat dehydration.

下列何者不是人畜共通傳染病病原？

(A) E 型肝炎病毒
(B) 豬流行性下痢病毒
(C) 日本腦炎病毒
(D) 第二型豬鏈球菌

下列有關豬流行性下痢之敘述，何者正確？

(A) 致病原為rotavirus
(B) 主要之臨床症狀為水樣下痢
(C) 主要病變集中在小腸及胃黏膜
(D) 小腸絨毛上皮細胞內可見嗜酸性核內包涵體

豬流行性下痢病毒屬於下列何種冠狀病毒？

(A) Alphacoronavirus
(B) Betacoronavirus
(C) Gammacoronavirus
(D) Deltacoronavirus

下列有關豬流行性下痢（porcine epidemic diarrhea）之敘述，何者錯誤？

(A) 病原為冠狀病毒
(B) 主要臨床症狀為血樣下痢
(C) 主要病變在小腸
(D) 未見結腸有組織病理變化

Hemagglutinating encephalomyelitis virus (vomiting and wasting disease)

Introduction

- First isolated from the brains of suckling pigs with **encephalomyelitis** in Canada in 1962.
- In 1969, an antigenically identical virus was isolated in England from suckling pigs showing **anorexia**, depression, **vomiting**, and **stunting**, but without signs of ~~encephalomyelitis~~.
- Worldwide distributed and widespread among swine.
- Mostly **subclinical**.

Etiology

Hemagglutinating encephalomyelitis virus (pHEV)

- Genus *Betacoronavirus*, subfamily of *Coronaviridae*
- Ss, positive-sense RNA genome of about 30 kb in length.
- One serotype
- The virus agglutinates erythrocytes of mice, rats, chickens, and several other animals.
- Pig is the natural host with strong **neural tropism.**

Transmission

- pHEV is excreted oronasally.
- Transmission can occur via nasal secretions, via nose-to-nose contact, and aerogenically
- No carrier of pHEV

Pathogenesis

- To develop clinical disease:

  -- The pig must become infected oronasally **prior to 3–4 weeks of age**

  -- The pig must originate from **nonimmune mothers**.

- Maternal Abs against pHEV prevent the virus from reaching neural target tissues and protect pigs from clinically affected.

- The type and severity of clinical signs vary and are related to **age**, possible differences in **virus virulence**, and the **course of viral pathogenesis**.

Pathogenesis

- pHEV replicate in the primary sites (either respiratory or GI tracts) and then spread to PNS → CNS via distinct pathways.

  -- Nasal mucosa → tonsil → trigeminal ganglion → trigeminal sensory nucleus in the brainstem

  -- Vagal nerve → vagal ganglion → vagal sensory nucleus in the brainstem

  -- Intestinal plexuses → local sensory ganglia →spinal cord

- Viremia is of little or no importance.
- Causes of vomiting: central, sensory, or GI

Clinical Presentation

- Varied morbidity and mortality
- Mostly subclinical.

The status of neonatal piglets

- Start with sneezing or coughing and fever.
- Two main clinical manifestations associated with pHEV neurotropism:

Repeated retching & vomiting → wasting

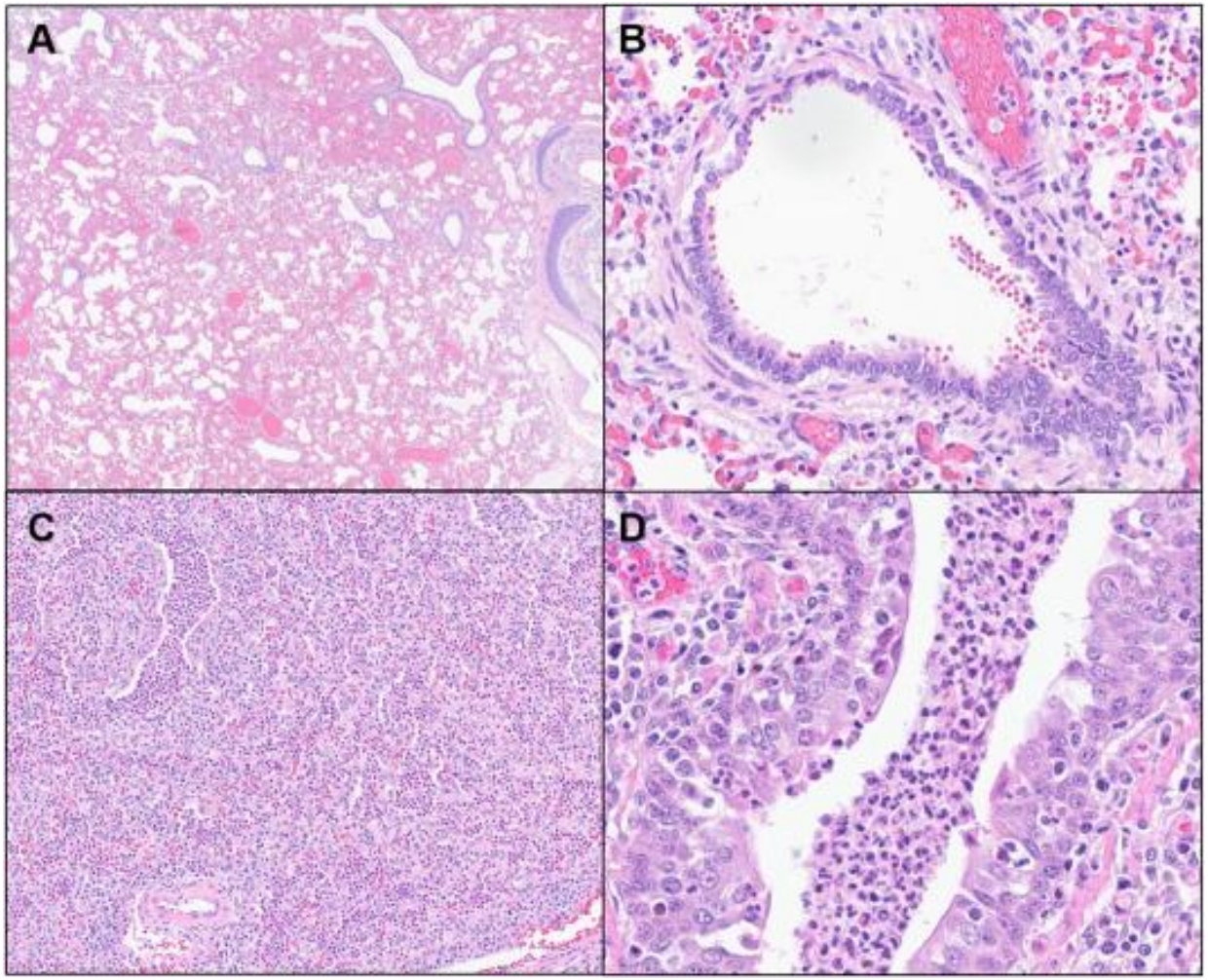
-- Typical VWD with frequent vomiting leading to death or subsequent wasting

-- Acute encephalomyelitis with motor disorders

-- Both manifestation may concur in the same outbreak

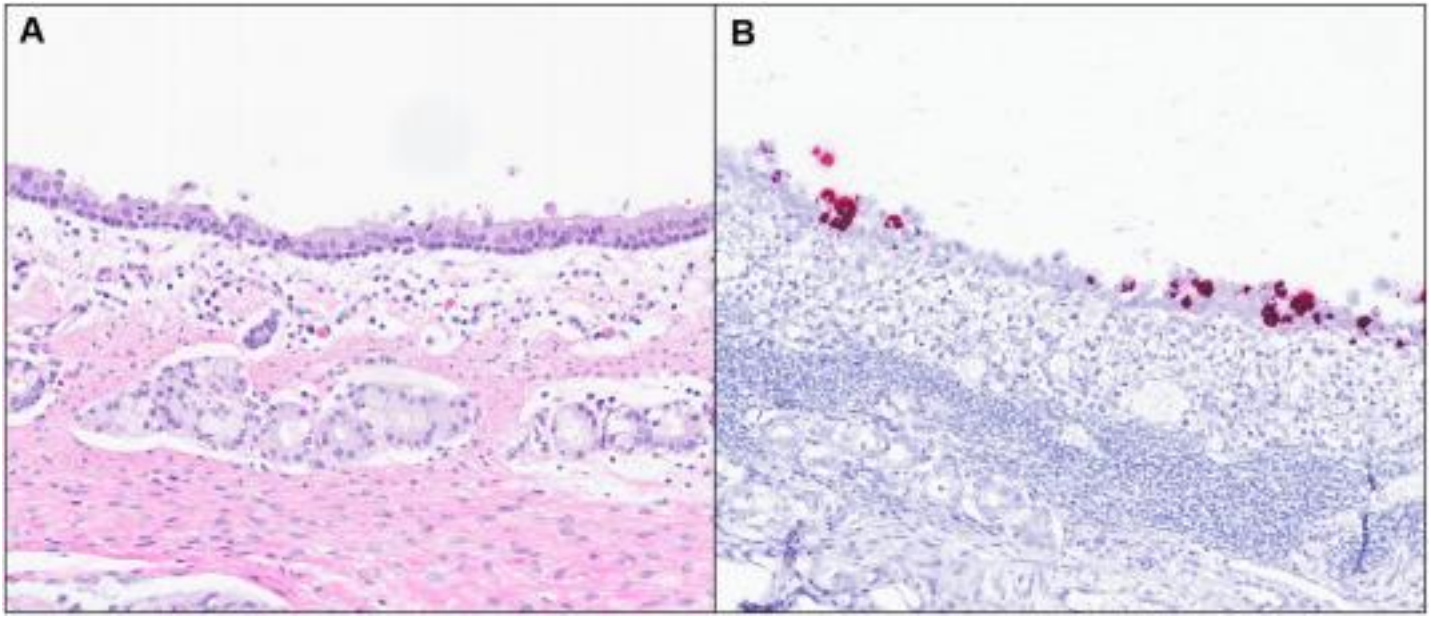
Lesions

- Gross: cachexia, stomach dilatation, distention of the abdomen
- Micro:
  - ✓Epithelial degeneration and inflammation in respiratory tract and tonsils
  - ✓Non-suppurative encephalomyelitis
  - ✓Degeneration of the ganglia of the stomach wall and perivascular cuffing



Histological changes associated with PHEV infection. Sections of lung with varying degrees of diffuse capillary congestion and attenuating and necrotizing bronchiolitis (Fig. 1A and B). There is also diffuse interstitial pneumonia characterized by lymphocytes and plasma cells expanding the alveolar septa (Fig. 1C). Bronchioles are lined by attenuated and hyperplastic epithelium and lumens contain sloughed necrotic debris and degenerative neutrophils (Fig. 1D).

Arunsiripate et al., 2025



Hematoxylin and eosin staining of trachea from a pig infected with PHEV. There is attenuation of the pseudocolumnar epithelium with squamous epithelial metaplasia. B) PHEV mRNA chromogenic ISH (RNAScope®) signal within the tracheal respiratory epithelium.

Arunsiripate et al., 2025

Laboratory Diagnosis

- Virus detection (feces or lesions)
  - ✓ Identification of infectious) virion: IEM, virus isolation

    *Tonsils, brainstem and lungs

    *Successful isolation can be checked by syncytia formation, **hemadsorption,** or **hemagglutination**.
  - ✓Identification of antigen: ELISA, IF, IHC
  - ✓Identification of nucleic acids: RT-PCR, RT-qPCR
- Serological test (oral, fecal, blood)
  - ✓ELISA
  - ✓Virus neutralization

Prevention & Control

- Promoting virus circulation in the farm so that gilts are immune at farrowing prevents disease in piglets

- No vaccines against pHEV are currently available.

21.下列有關血球凝集腦脊髓炎（hemagglutinating encephalomyelitis）之敘述，何者錯誤？

A.病原主要感染3週齡以下的豬
B.病原只有一個血清型
C.病原會引起嘔吐衰弱
D.病原與日本腦炎病毒類似

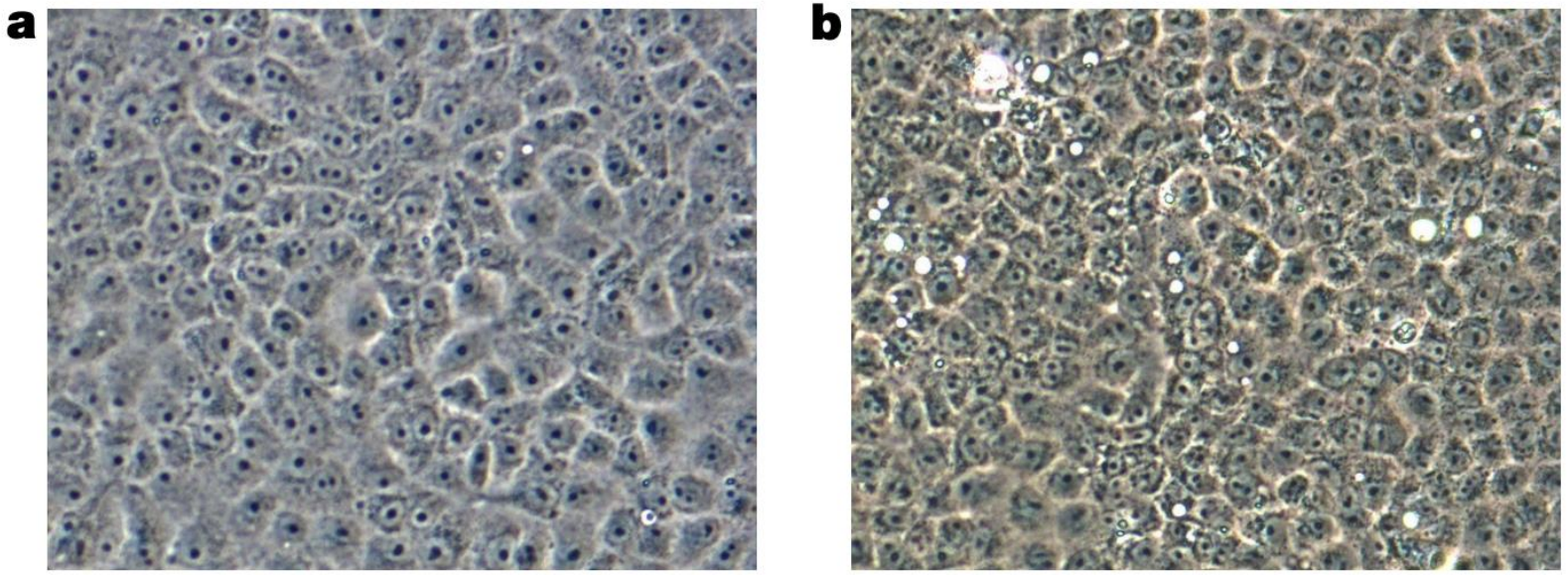
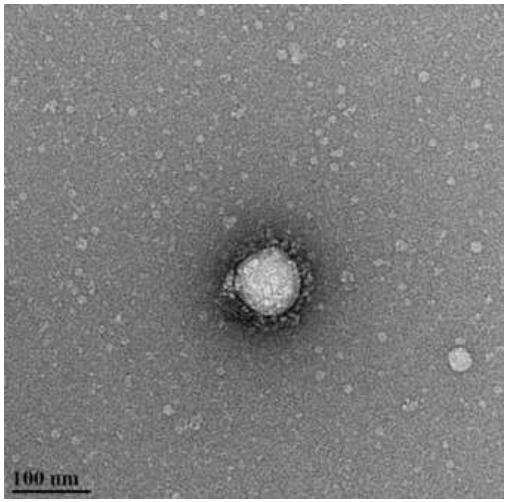


Fig. 1 | Non-specific CPEs formed by plasma from patient 0081-4 in Vero E6 cells. a, Mock-infected Vero E6 cells, 11 days post-inoculation with phosphate-buffered saline. **b**, Vero E6 cells 11 days post-inoculation with plasma from patient 0081-4. Original magnification, ×200.

Three unbiased approaches:

- Viral isolation using Vero E6 cells
- PCR
- EM



台灣PEDV案例分享

豬流行性下痢蔓延 防檢局研商對策

#防檢局 #小豬 #感染 #法定傳染病 ...

發布時間：2014/2/15 18:41 更新時間：2023/1/13 17:50

農曆年前，南部爆發豬下痢疫情，現在已經蔓延到中北部地區，根據農委會防檢局統計，過去三個月內，全台總共有12萬7千多頭小豬仔，疑似被感染死亡! 工作人員在養豬場忙著噴藥消毒，因為最近豬圈內的小豬，因感染「豬流行性下痢」，病死了很多小豬仔，而根據農委會防檢局統計，最近三個月以來，全台共有12萬7千多頭小豬，疑似感染豬下痢死亡，是歷年來最嚴重的一次疫情。 ==防檢局副局長 趙磐華== 它是屬於一種高致死率的 就是高傳染性 特別是高傳染性致死率的病毒 跟過去的幾年不同 確實是變異 防檢局表示，「豬流行性下痢」是一種冠狀病毒，好發於冬末春初，母豬感染後會出現食慾不振、減少泌乳量，但過一陣子就會產生抗體、自然痊癒，不過剛出生的小豬，因為沒什麼抵抗力，一旦感染會嚴重脫水，出生不到14天就會死亡，而豬農也擔心，小豬養不活，到端午節時、恐怕沒有溫體豬可賣，價格可能再飆漲。 ==聲音來源：養豬協會常務監事 潘連周== 目前的豬價 每百公斤七千四百塊 （那平常呢 平常的價格是） 六千幾啊 六千七 六千七 六千八而已啊 根據養豬協會統計，屏東是養豬大縣，而且有八成豬場都受到感染，因此推估全台至少有30萬頭豬死亡，而防檢局則表示，下週一、17號將召開學者專家會議，討論是否把豬下痢改為法定傳染病、必須要通報，另外也考慮研發本土型疫苗、加強防疫。 記者 賴淑敏 蔣龍祥 台北報導

#統計 #死亡 #農委會 #冠狀病毒

https://news.pts.org.tw/article/262158

病例2014

張文發 副主任
中興大學獸醫學院動物疾病診斷中心

病史

- 一貫場
- 該場在養3,000頭
- 10日齡豬隻出現瘦弱、下痢(水樣、糊狀)及嘔吐等症狀
- 發病率: 23%（60/260)
- 死亡率: 50%
- 2014，1月送5頭10日齡發病仔豬檢驗

病理檢驗 _ 外觀





病理解剖



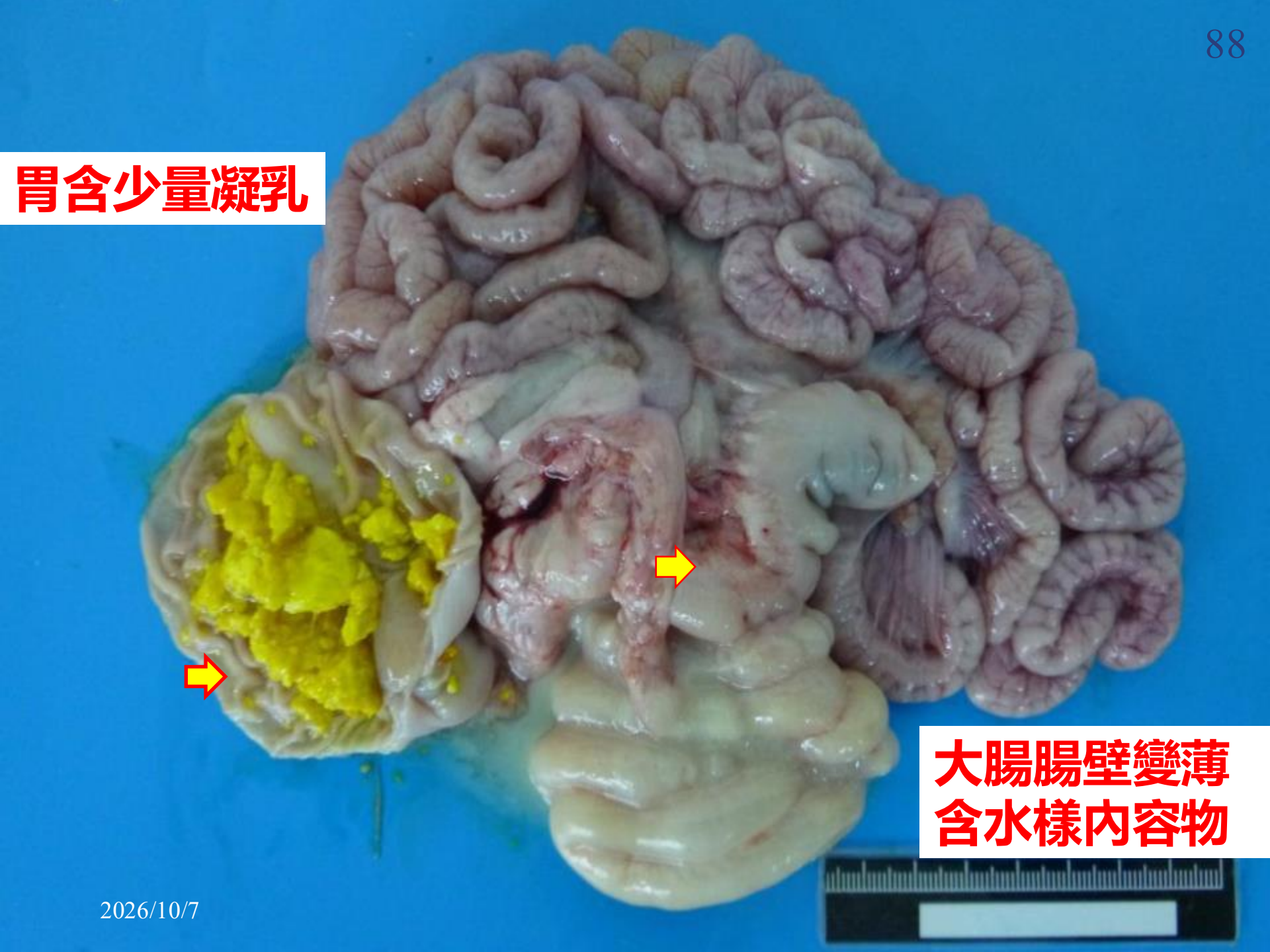


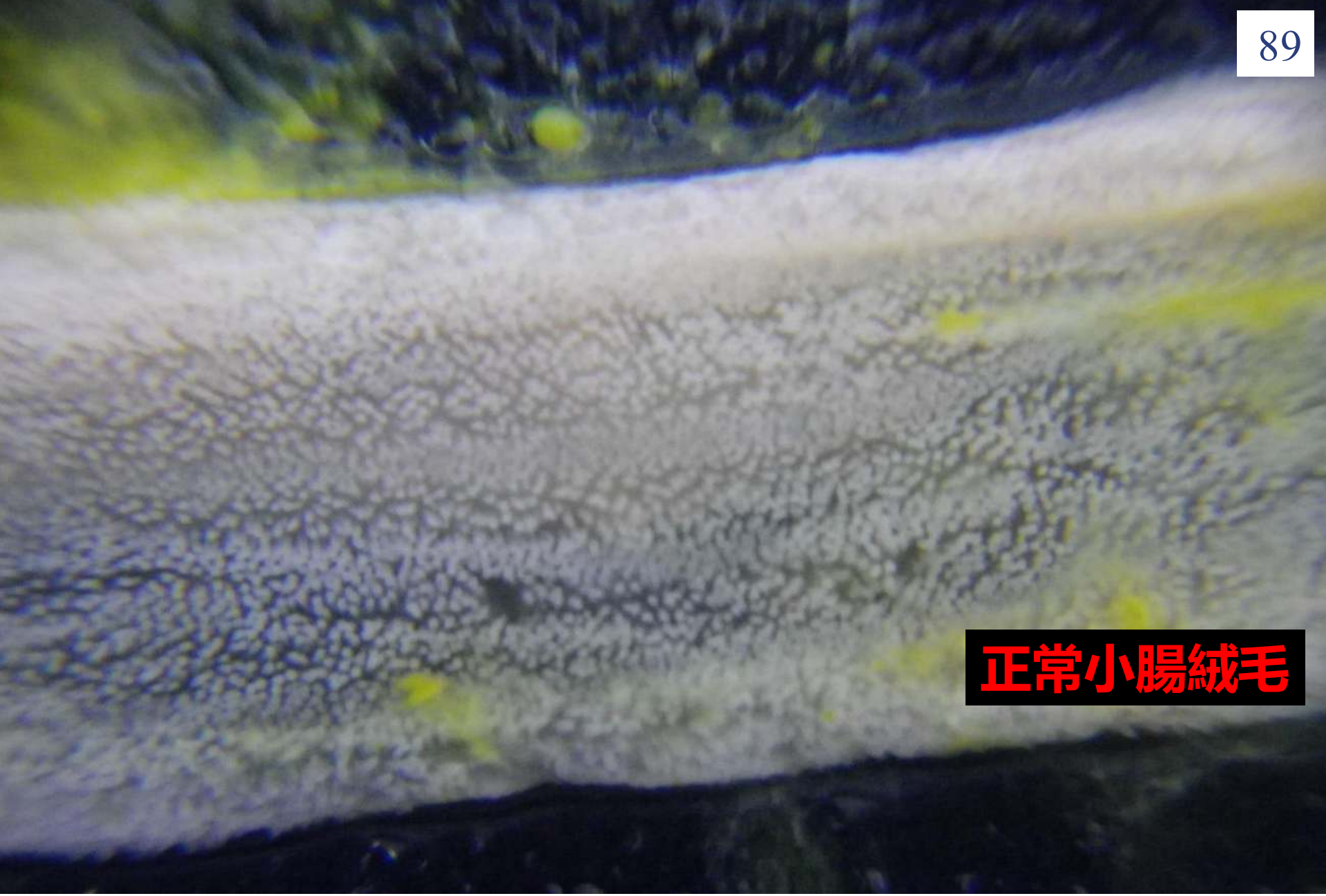
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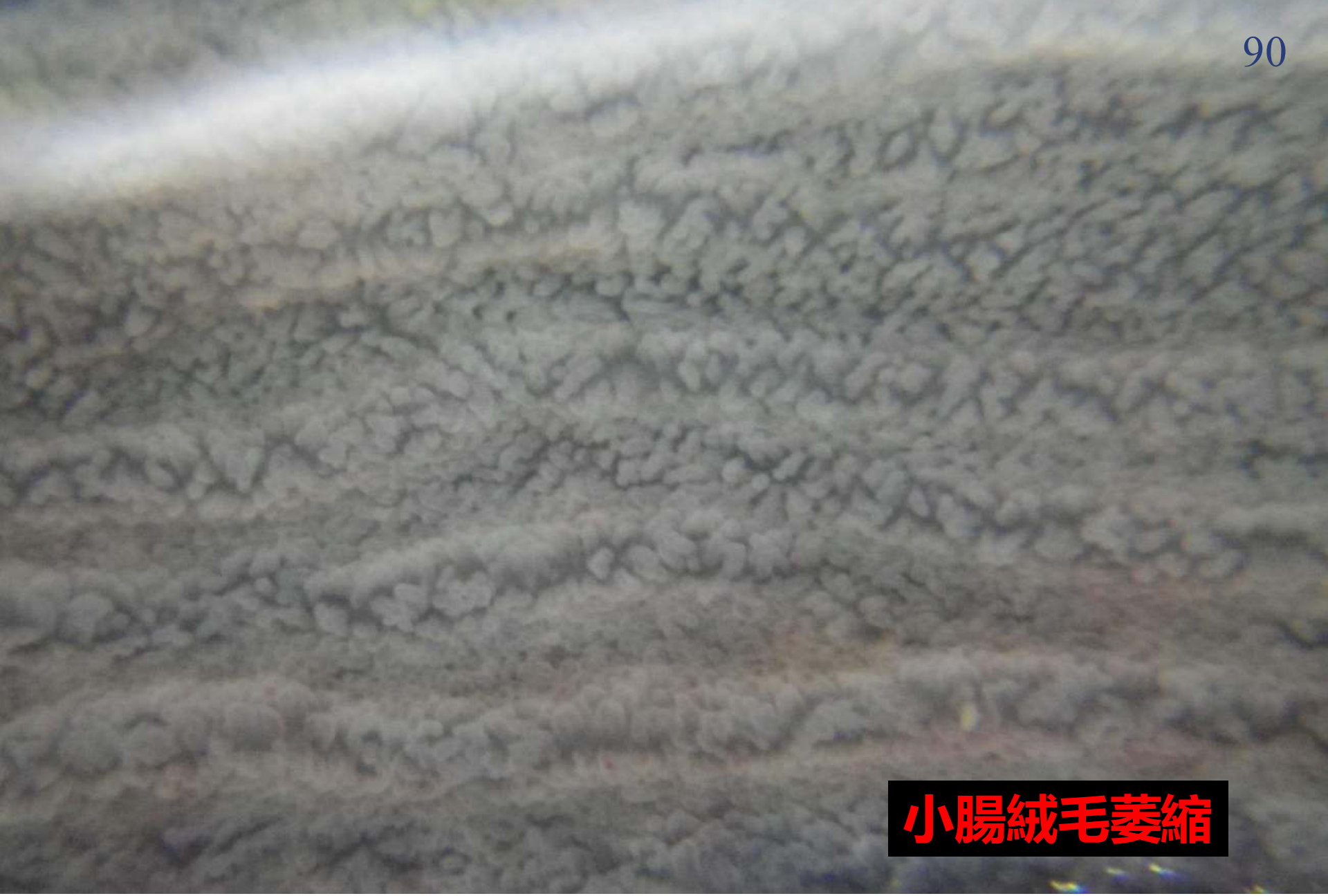

乳糜管丧失功能



空腸腸繫膜淋巴結腫大


胃含少量凝乳
大腸腸壁變薄
含水樣內容物


正常小腸絨毛


小腸絨毛萎縮

初步診斷/Tentative diagnosis

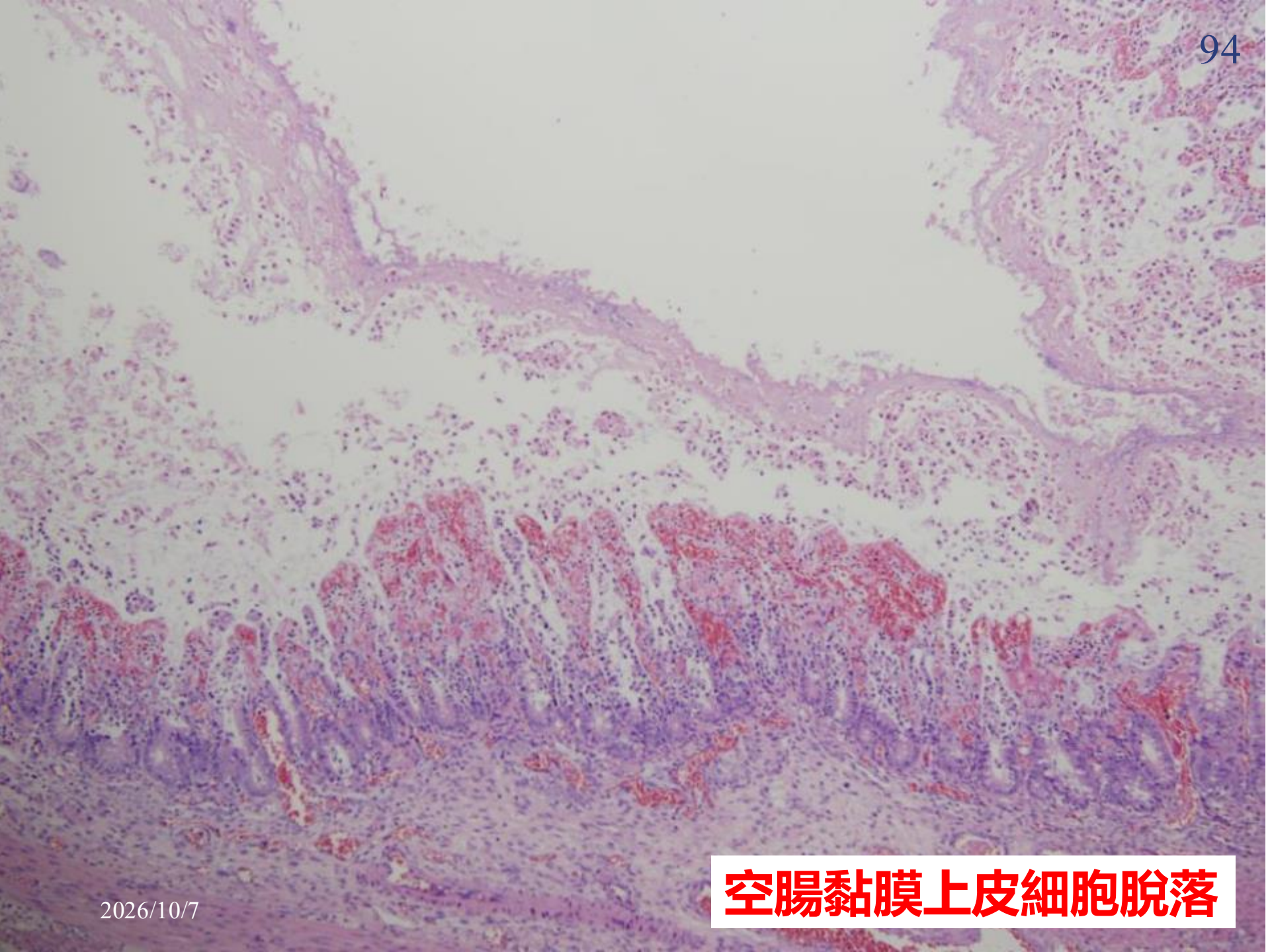
哺乳豬腸炎

組織病理檢驗 (H & E stain)





空腸絨毛萎縮

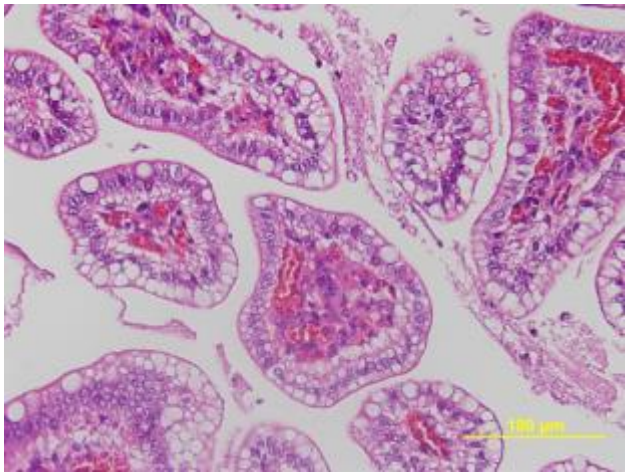


空腸黏膜上皮細胞脫落

免疫螢光檢驗

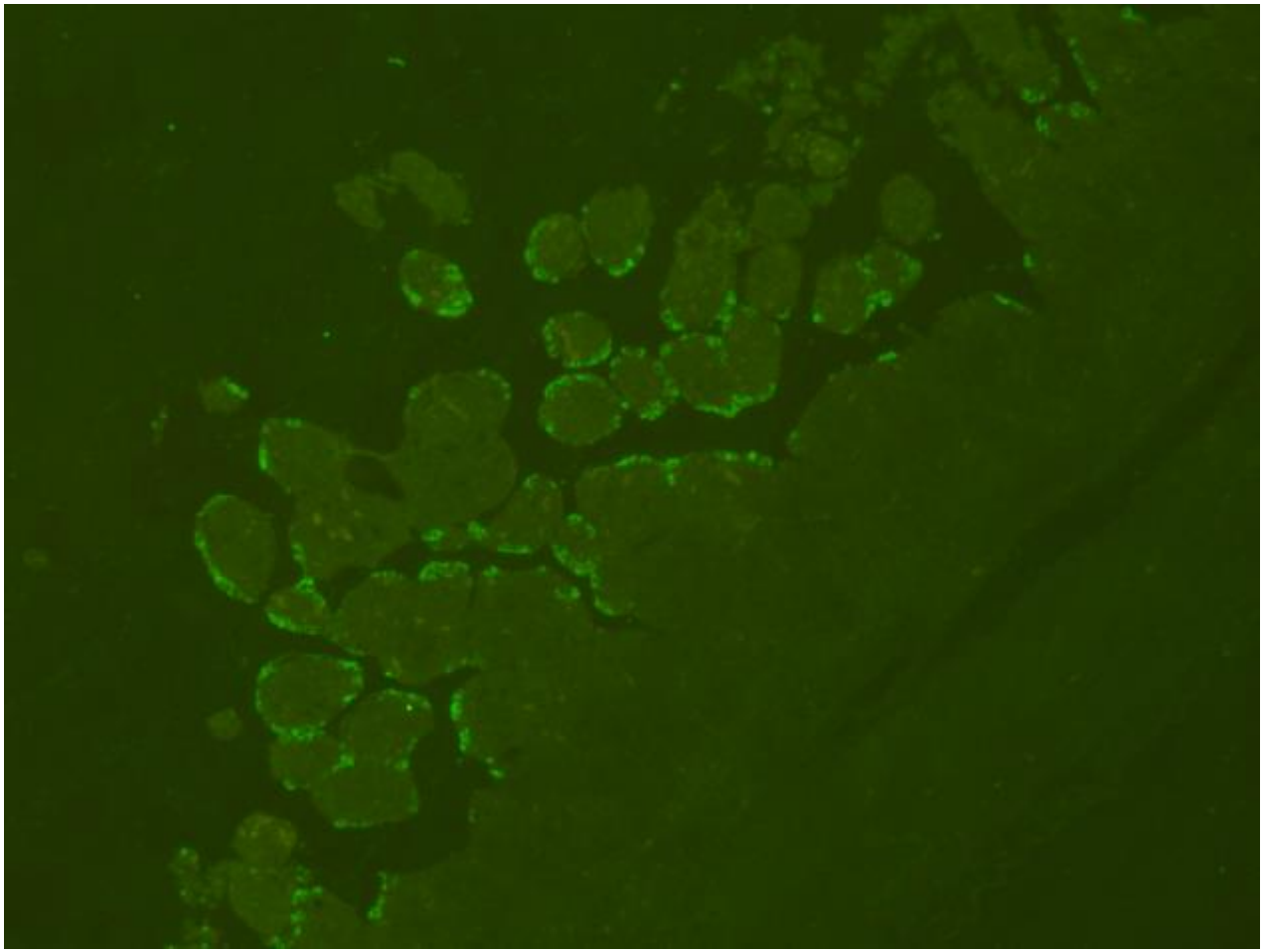
MAb

Immunofluorescent staining



豬流行性下痢病毒
PEDV

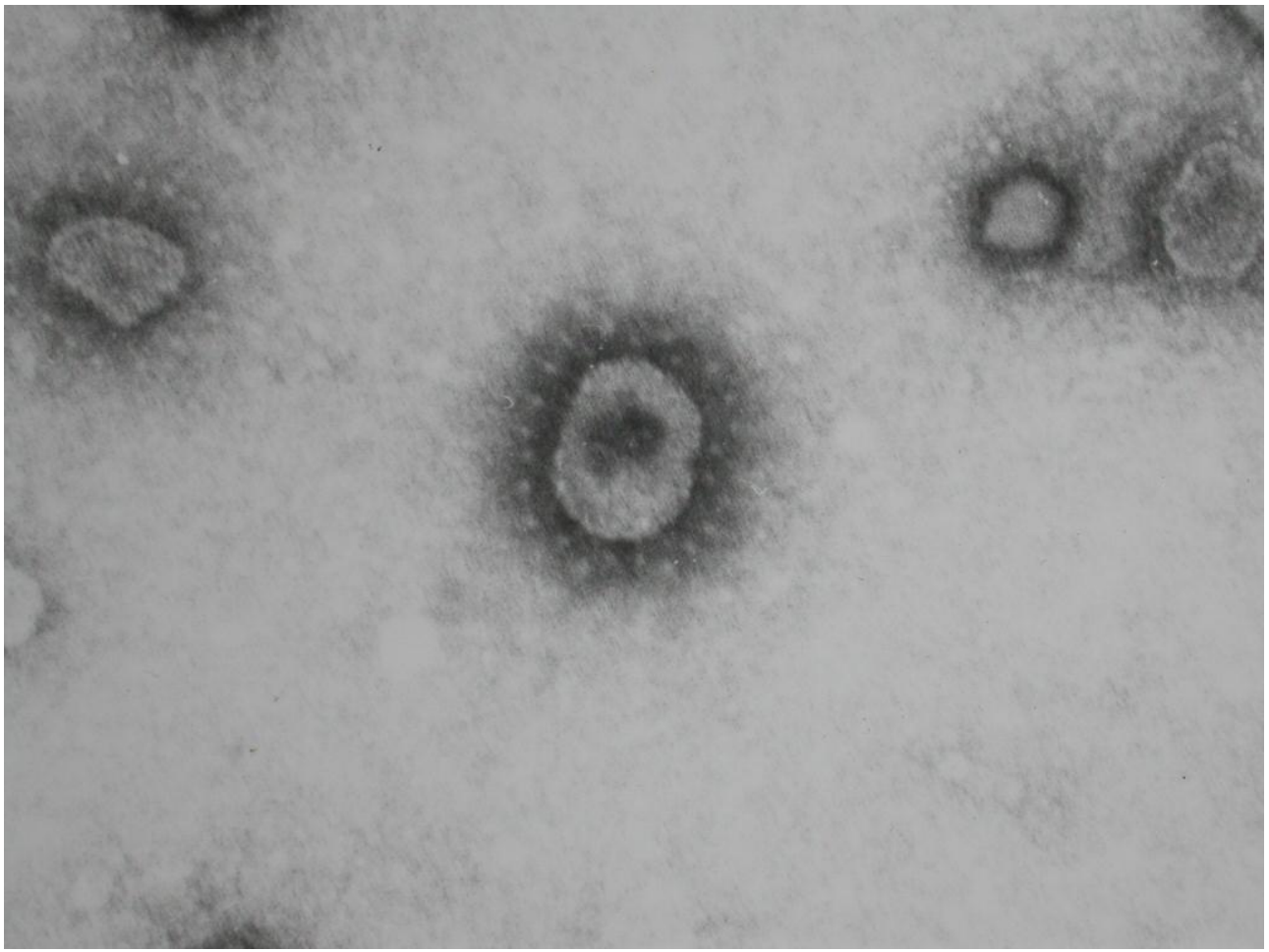
Immunofluorescent staining



其它實驗室檢驗

- 電子顯微鏡
- 分生 (RT-PCR)
- 細菌分離

電子顯微鏡檢查

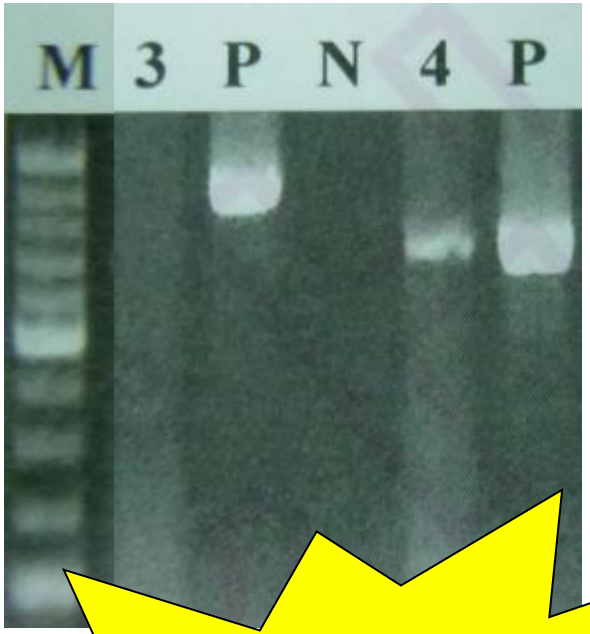


穿透式電子顯微鏡檢查可見冠狀病毒顆粒

Reverse transcription polymerase chain reaction

RT-PCR for TGEV & PEDV

small intestines



PEDV: 691 bp

M: DNA marker
3: Sample for TGEV
4: Sample for PEDV
P: Positive control
N: Negative control

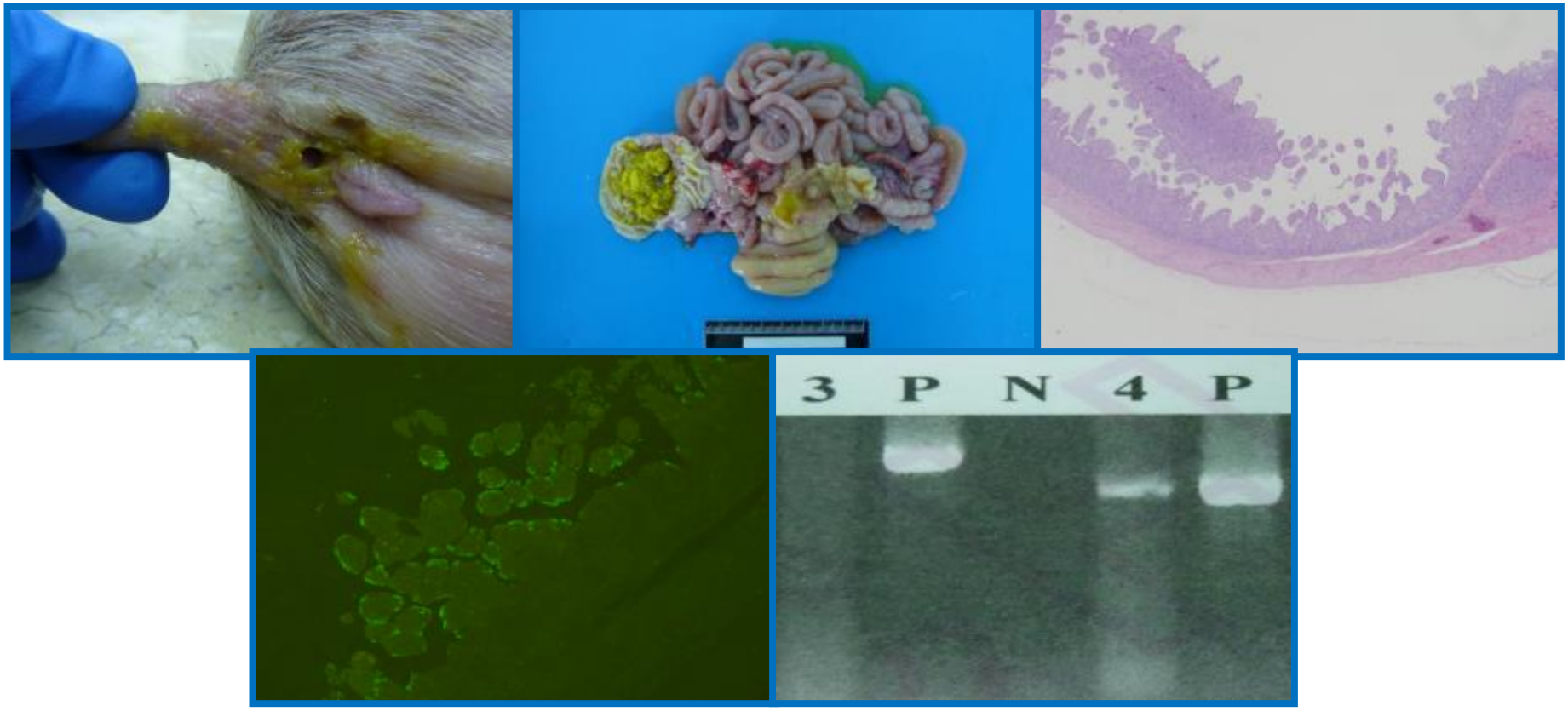
豬流行性下痢(PEDV)　陽性
傳染性胃腸炎(TGEV)　陰性

小腸釣菌培養



抗藥物敏感性試驗

Item	(*Escherichia coli*)
Amoxicillin-Clavulanic acid	I
Ampicillin	R
Nalidixic acid	R
Ceftiofur	I
Colistin	R
Doxycycline	R
Cephalexin	S
Enrofloxacin	S
Flofenicol	S
Streptomycin	R
Lincomycin/spectinomycin	R
Cefquinome	S



診斷/Final diagnosis

豬流行性下痢(PED)

未來之對策

1. 快速確診，~~快速反飼(*feedback*)~~可降低損失



2. 地區性流行、四季發生

   (如U.S)

3. 一旦發病，仔豬死亡率較低

4. 防疫(Biosecurity)不佳之豬場為源頭

5. 自家疫苗效力受制於保存的條件

Porcine Deltacoronavirus (PDCoV)

Introduction

- A newly emerging swine enteric viral disease, caused by porcine deltacoronavirus (PDCoV).
- Probably as a consequence of one or more recombination events involving avian deltacoronaviruses
- Share **similar but less severe** clinical and pathological features with those caused by TGE and PEDV.
- PDCoV exhibit **broad host range** and may be incompletely adapted to pigs → spread more slowly among pigs comparing to PEDV.
- ***Spillover to human?***
- Diagnostic approaches, immunity and prevention measurement are similar to those described for TGEV and PEDV.

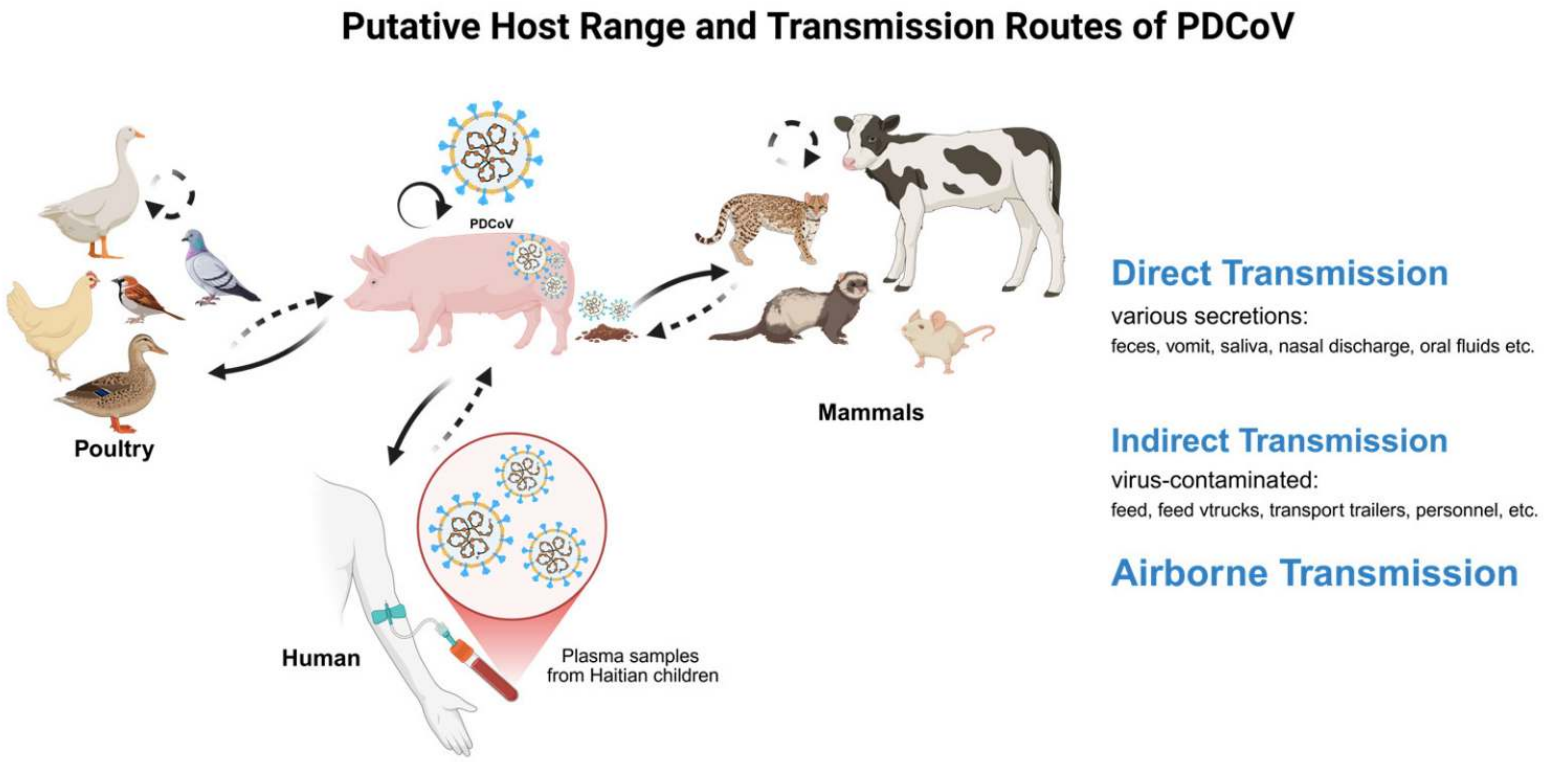


Figure 3. Schematic illustration of putative PDCoV host range and transmission routes. PDCoV infects both poultry and mammalian hosts, primarily through three possible epidemiological routes: direct contact, indirect fomite-mediated, and airborne aerosol transmission. Solid arrows denote transmission events confirmed experimentally in animal models, at the cellular level, or supported by epidemiological evidence; dashed arrows indicate potential or inferred transmission routes that have not been definitively documented but cannot be excluded. https://BioRender.com/rgu32tu, accessed on 15 September 2026.

Article

Independent infections of porcine deltacoronavirus among Haitian children

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Coronaviruses have caused three major epidemics since 2003, including the ongoing SARS-CoV-2 pandemic. In each case, the emergence of coronavirus in our species has been associated with zoonotic transmissions from animal reservoirs[1,2], underscoring how prone such pathogens are to spill over and adapt to new species. Among the four recognized genera of the family *Coronaviridae*, human infections reported so far have been limited to alphacoronaviruses and betacoronaviruses[3–5]. Here we identify porcine deltacoronavirus strains in plasma samples of three Haitian children with acute undifferentiated febrile illness. Genomic and evolutionary analyses reveal that human infections were the result of at least two independent zoonoses of distinct viral lineages that acquired the same mutational signature in the genes encoding Nsp15 and the spike glycoprotein. In particular, structural analysis predicts that one of the changes in the spike S1 subunit, which contains the receptor-binding domain, may affect the flexibility of the protein and its binding to the host cell receptor. Our findings highlight the potential for evolutionary change and adaptation leading to human infections by coronaviruses outside of the previously recognized human-associated coronavirus groups, particularly in settings where there may be close human–animal contact.

Swine Acute Diarrhea Syndrome coronavirus (SADS-CoV)

Introduction

- A newly emerging swine enteric viral disease belonging to the Genus *alphacoronavirus*
- First detected in 2017 during a devastating enteric disease outbreak in southern China; currently a regional swine enteric disease
- High similarity (>95%) between SADS-CoV and **bat** CoV HKU2
- Presumably transmitted via fecal–oral route
- ***Broad host range. Zoonotic potential?***
- Rely on a more conserved host factor that could contribute to its broad cell tropism and cross-species infec- tivity